



Functional Dirichlet Series, Integral Representations of a Generalized Zeta Function, and Applications to Sustainability

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Contents

1	Introduction	392
2	Background and Fundamental Definitions	396
3	Sequence and Series Convergences	401
3.1	Sequence Convergences	401
3.2	Series Convergences	406
4	Integral Representations	413
4.1	Interchange of Integration and Summation	413
4.2	Integral Representation and Explicit Formulas	419
4.3	Connection to the Harmonic Series	422
4.4	Special Case: $s = 2$ and the Basel Problem	427
5	Applications	429
5.1	Harmonic Numbers for a Greener World	429
5.2	Generalization: Optimal Resource Allocation with Varying Efficiency Decay	431
5.3	Emerging Applications in Sustainability	433
6	Conclusion	434
7	Cross-References	436
	References	436

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Abstract

We investigate a generalized Dirichlet series with functional coefficients involving products of exponential and hyperbolic secant functions, extending our previous work on harmonic series representations. By introducing a sequence of functions $f_n(s, x)$ parameterized by a complex number s with $\operatorname{Re}(s) > 1$ and real number $x \geq 0$, we establish both pointwise and uniform convergence properties of the associated series $\sum_{n=1}^{\infty} f_n(s, x)$. We verify that although the series converges pointwise on the entire domain $E = \{(s, x) : \operatorname{Re}(s) > 1, x \geq 0\}$, uniform convergence fails on E but holds on appropriately restricted subsets, leading to locally uniform convergence on compact subsets. The sum function is expressed explicitly in terms of polylogarithm functions and the Riemann zeta function. Through integral representations obtained via term-by-term integration, we derive connections to classical results, that is, as $s \rightarrow 1^+$, our framework provides an alternative proof of the divergence of the harmonic series, while the case $s = 2$ recovers Euler's solution to the Basel problem, $\zeta(2) = \pi^2/6$. We characterize convergence rates on restricted domains, showing exponential decay $O(r_\varepsilon^n)$ when x is bounded away from zero and polynomial decay $O(1/n^{1+\delta})$ when $\operatorname{Re}(s)$ is bounded away from the critical value 1. Finally, we explore potential applications to sustainability through a work-rate model illustrating diminishing returns in resource allocation and discuss emerging theoretical connections between the Riemann zeta function and environmental modeling in complex systems, climate dynamics, and extreme event analysis.

Keywords

Functional Dirichlet series · Generalized Riemann zeta function · Integral representations · Polylogarithm functions · Harmonic series · Sustainability applications

1 Introduction

In mathematics, the harmonic series is an infinite sum where the terms are the reciprocals of the positive integers, given as follows:

$$\sum_{n=1}^{\infty} \frac{1}{n} = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots$$

Its partial sum, the sum of the first n terms, is called the n -th harmonic number H_n and is expressed as (Graham et al. 1994)

$$H_n = \sum_{k=1}^n \frac{1}{k} = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots + \frac{1}{n}.$$

The harmonic series is often introduced in calculus when discussing sequences and series and their convergence properties. Most freshman-level calculus textbooks usually present one or two proofs of its divergence (Anton et al. 2021; Stewart et al. 2020; Thomas et al. 2022). Historically, it was the French mathematician Nicole Oresme (circa 1323–1382) who first demonstrated in the middle of the fourteenth century that the harmonic series diverges. His simple and elegant proof uses the comparison test, where each denominator in the series is replaced with the next largest power of 2. The technique involves grouping terms of the harmonic series and showing that although the individual terms decrease, the sum of each group remains bounded below by a positive constant, causing the total sum to diverge (Bell and Blåsjö 2018).

Although Oresme's manuscript was not published until the twentieth century, the first known published proof of harmonic series' divergence was written by Pietro Mengoli in 1650 (Mengoli 1650). While Oresme's proof was based on grouping the terms of the series into blocks of 2^n , $n \in \mathbb{N}$, Mengoli grouped the terms into blocks of 3. He argued that the first term exceeds the middle term by more than the middle one exceeds the last term, that is,

$$\frac{1}{n-1} - \frac{1}{n} > \frac{1}{n} - \frac{1}{n+1},$$

for which he obtained and applied the following inequality (Mengoli 1650; Bell and Blåsjö 2018):

$$\frac{1}{n-1} + \frac{1}{n} + \frac{1}{n+1} > \frac{3}{n}.$$

A standard calculus approach to demonstrating divergence uses the integral test by comparing the harmonic series with the integral of the function $1/x$:

$$\int_1^\infty \frac{1}{x} dx,$$

which is a divergent improper integral. In fact, there are several dozen known proofs that the harmonic series diverges (Bonar and Khoury 2006; Kifowit and Stamps 2006). See also Baker (2015), Clarkson (1966), Cusumano (1998), Goldmakher (2022), Kifowit (2006) and Ward (1970) for additional proofs of the divergence of the harmonic series.

A fundamental relationship exists between the harmonic series and the Euler–Mascheroni constant (Havil 2017; Knuth 1997). Often denoted as γ , this constant is defined by the limiting difference between the harmonic number H_n and the natural logarithm as the number of terms approaches infinity:

$$\gamma = \lim_{n \rightarrow \infty} (H_n - \ln n).$$

Its approximate value is $\gamma \approx 0.5772156649$. It remains an open question whether the Euler-Mascheroni constant is irrational, let alone transcendental (i.e., not the root of any nonzero polynomial with rational coefficients) (Brent 1977; Dence and Dence 2009; Lagarias 2013; Olaikhan 2021). The Euler–Mascheroni constant appears in diverse applications, from modeling atmospheric boundary layers to quantum information theory (Owinoh et al. 2005; Caves and Fuchs 1996).

The harmonic series is a special case of the Riemann zeta function $\zeta(s)$ when $s = 1$ (Apostol 2013; Miller and Takloo-Bighash 2006). Conversely, the Riemann zeta function can be viewed as a generalization of the harmonic series where each term is raised to a complex power $s \in \mathbb{C}$:

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s} = 1 + \frac{1}{2^s} + \frac{1}{3^s} + \cdots.$$

Although the harmonic series diverges, this series converges to a finite value when the real part of s is greater than 1.

A particularly celebrated special case occurs when $s = 2$, reducing to the famous Basel problem. This problem was proposed in 1650 by Pietro Mengoli and solved in 1734 by Leonhard Euler (Dunham 1999; Hollingdale 2011; Sandifer 2007):

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

Readers interested in exploring the Basel problem further, including its historical context and significance within number theory, may consult Derbyshire (2004), Merzbach and Boyer (2011), and Weil (2007).

Although the Riemann zeta function is defined and converges for $s \in \mathbb{C}$ with $\operatorname{Re}(s) > 1$, it can be extended to the entire complex plane through analytic continuation. This extended function is a meromorphic function with a simple pole at $s = 1$ (Bateman 2004; Miller and Takloo-Bighash 2006).

The Riemann zeta function has a fundamental connection to prime numbers, first noticed by Leonhard Euler in the 1730s and published in 1737 in the inaugural journal series of the St. Petersburg Academy of Sciences (Euler 1737). This connection is expressed through the Euler product formula, which equates the Riemann zeta function with an infinite product over all primes p (Erickson and Vazzana 2008; Miller and Takloo-Bighash 2006; Patterson 1995):

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s} = \prod_{p \text{ prime}} \frac{1}{1 - p^{-s}}.$$

This formula is a fundamental result in analytic number theory because it establishes a deep connection between the continuous analysis of the Riemann zeta function and the discrete number theory of prime numbers (Hardy and Wright 2008; Jameson 2008; Rempe-Gillen et al. 2014).

The Riemann zeta function has two types of zeros: trivial and nontrivial. The trivial zeros arise from the functional equation of the zeta function and occur at the negative even integers $s = -2n$ for $n \in \mathbb{N}$. The nontrivial zeros lie within the critical strip $0 < \operatorname{Re}(s) < 1$. In a groundbreaking paper published in 1859, Bernhard Riemann conjectured that all nontrivial zeros lie on the vertical line $\operatorname{Re}(s) = 1/2$ in the complex plane, now called the *critical line* (Riemann 1859). This conjecture remains unproven and is known as the Riemann hypothesis.

Edwards (1974) has written a highly regarded book on Riemann's zeta function that examines Riemann's landmark 1859 paper and subsequent developments in number theory, including an English translation of Riemann's original paper in the Appendix. For readers seeking a more accessible introduction to prime numbers and the Riemann hypothesis, Mazur and Stein (2016) provide an excellent resource requiring minimal mathematical background. For a narrative account of the ongoing search for a proof, Rockmore (2005) chronicles the historical and interdisciplinary dimensions of the Riemann hypothesis, including its connection to quantum physics.

For more advanced expositions, Titchmarsh (1986)'s book is considered an all-time classic. Widely cited and highly regarded by experts in the field, it provides a foundational text for the study of analytic number theory and classical analysis. Other comprehensive texts covering the theory and applications of the Riemann zeta function include Ivić (2003) and Patterson (1995). See also Montgomery (2017) for historical perspectives on the Riemann zeta function and an exploration of both classical and modern results in analytic number theory, harmonic analysis, complex analysis, and probability theory, Srivastava and Choi (2012) for various integral representations associated with zeta functions and related functions, and Arakawa et al. (2014) for a comprehensive treatment of the relationship between the zeta function and Bernoulli numbers.

In this study, we develop a generalized framework involving the polylogarithm function and Dirichlet series with functional coefficients. By considering products of exponential and hyperbolic secant functions—a combination that naturally appears in soliton theory—we establish connections to fundamental results including the divergence of the harmonic series and the Basel problem. Our work is related to Varadarajan's foundational contributions to the analytic theory of Dirichlet series, particularly regarding functional equations and integral representations of generalized zeta functions. In his seminal paper, Varadarajan (2007) traced the historical development of functional equations from Euler's transformation formulas through Riemann's analytic continuation to modern L -function theory.

Our primary contribution is methodological. We demonstrate how traditional Eulerian techniques of series manipulation, combined with rigorous convergence analysis and integral representations, provide alternative proofs for fundamental results in analytic number theory. Specifically, we establish new pathways to prove (a) the divergence of the harmonic series and (b) Euler's solution to the Basel

problem, both emerging as limiting cases of our generalized framework. While the results themselves are classical (the divergence of $\sum 1/n$ and the value $\zeta(2) = \pi^2/6$), our proofs are novel in their use of functional coefficients, interchange theorems for integration and summation, and limiting processes as parameters approach critical values. This work complements existing approaches by providing a physics-inspired perspective grounded in the analytical properties of functions arising from nonlinear partial differential equations.

Our framework belongs to the broader study of Dirichlet series with functional coefficients, distinct from but related to the modern algebraic theory of multiple zeta values (MZVs). While we do not directly employ shuffle or stuffle product structure characteristic of MZV theory, our approach shares conceptual ground with classical analytic methods: term-by-term manipulation, convergence analysis at critical boundaries, and integral representations. The polylogarithm functions $\text{Li}_s(z)$ that emerge naturally in our formulas (see Theorem 2, equation (5)) are well-studied objects in MZV theory, and our explicit representations may offer complementary perspectives to purely algebraic reduction techniques.

This work is organized as follows. Section 2 provides background and fundamental definitions. Section 3 establishes convergence properties for both sequences and series, examining pointwise and uniform convergence. Section 4 develops integral representations of the series and derives explicit formulas, making connections to the harmonic series and the Basel problem. Section 5 explores potential applications to sustainability science. Finally, Section 6 concludes with a summary of results and directions for future research.

2 Background and Fundamental Definitions

Drawing upon our previous work, which related harmonic series to integrals involving combinations of exponential and hyperbolic secant functions (Karjanto 2015, 2016, 2021), we extend a similar approach to include a generalization of the Riemann zeta function and the Dirichlet series with functional coefficients. We begin with the following definitions commonly found in the mathematics literature.

Definition 1 (Riemann Zeta Function). For $s \in \mathbb{C}$ with $\text{Re}(s) > 1$, the *Riemann zeta function* is given by the following infinite series that converges absolutely to an analytic function (Erickson and Vazzana 2008; Miller and Takloo-Bighash 2006; Patterson 1995; Pongsriam 2023):

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s} = \frac{1}{1^s} + \frac{1}{2^s} + \frac{1}{3^s} + \frac{1}{4^s} + \cdots \quad (1)$$

This function, extensively studied by Bernhard Riemann in his famous 1859 paper, is central to number theory and is particularly famous for its connection to the distribution of prime numbers and for the unsolved Riemann hypothesis (Riemann

1859; Edwards 1974). For $s = 1$, the associated series reduces to the harmonic series, which diverges to $+\infty$. Furthermore, it has a simple pole with residue 1, which means that it behaves like $1/(s - 1)$ near $s = 1$, that is,

$$\lim_{s \rightarrow 1} (s - 1)\zeta(s) = 1.$$

Definition 2 (Dirichlet Series). A *Dirichlet series* is an infinite series of the form (Apostol 2013; Hardy and Wright 2008; Miller and Takloo-Bighash 2006; Pongsriiam 2023):

$$h(s) = \sum_{n=1}^{\infty} \frac{a_n}{n^s},$$

where $s \in \mathbb{C}$ is a complex variable and the coefficients $\{a_n\}$ are a sequence of complex numbers.

The series converges in some right half plane of the complex plane, typically for $\operatorname{Re}(s) > \sigma_0$, where σ_0 is called the *abscissa of convergence*. We observe that the Riemann zeta function is a special case of a Dirichlet series where all coefficients $a_n = 1$.

Remark 1 (Convergence Abscissas). For Dirichlet series, three types of convergence can be distinguished: simple convergence (at abscissa σ_c), uniform convergence (at abscissa σ_b), and absolute convergence (at abscissa σ_a). It is known that $\sigma_c \leq \sigma_b \leq \sigma_a$ with $\sigma_a - \sigma_c \leq 1$ in general (Bohr 1913). When the coefficients possess multiplicative structure, these abscissas exhibit special relationships. In particular, Brevig and Heap (2016) proved that for Dirichlet series with multiplicative coefficients, the abscissas of uniform and absolute convergence coincide: $\sigma_a = \sigma_b$.

Definition 3 (Dirichlet Series with Functional Coefficients). For the purposes of this work, we define a *Dirichlet series with functional coefficients* as a series of the form

$$h(s, z) = \sum_{n=1}^{\infty} \frac{g_n(z)}{n^s},$$

where $s \in \mathbb{C}$, $z \in D \subseteq \mathbb{C}$, and each $g_n: D \rightarrow \mathbb{C}$ is a given complex-valued function, where the series converges in a specific region of the (s, z) -space.

Remark 2. A related but different generalization involves translated Dirichlet series. For a Dirichlet series $h(s) = \sum_{n=1}^{\infty} a_n n^{-s}$ and $\sigma \in \mathbb{R}$, the translated series

$h_\sigma(s) = \sum_{n=1}^{\infty} \frac{a_n}{n^{1/\sigma}} n^{-s}$ satisfies $h_\sigma(s) = h(s + 1/\sigma)$. Fernández Vidal et al. (2021)

studied the Hardy space H^+ of Dirichlet series whose translations belong to H^p for all $\sigma > 0$, establishing that this forms a Fréchet–Schwartz space with the Dirichlet monomials as an unconditional Schauder basis. While their translation operator acts on the complex variable s , our functional coefficients $g_n(z)$ depend on an auxiliary parameter z , providing a different type of generalization.

A particularly important special case of Dirichlet series with functional coefficients is the polylogarithm function, which plays a central role in our analysis.

Definition 4 (Polylogarithm or Jonquière’s Function). The *polylogarithm function*, denoted as $\text{Li}_s(z)$, is defined by the following power series in the variable z :

$$\text{Li}_s(z) = \sum_{n=1}^{\infty} \frac{z^n}{n^s}, \quad (2)$$

where a complex number z is the argument and s is the order or index, which can be any complex number (Lewin 1981).

We observe that the polylogarithm function (2) is a special case of the Dirichlet series with functional coefficients where $g_n(z) = z^n$. Furthermore, when $z = 1$ and $\text{Re}(s) > 1$, the polylogarithm function becomes the Riemann zeta function $\zeta(s)$ (1).

Theorem 1. The polylogarithm function $\text{Li}_s(z)$ (2) converges absolutely on the following domain D_p :

$$D_p = \{z \in \mathbb{C} : |z| < 1\} \cup \{z \in \mathbb{C} : |z| = 1 \text{ and } \text{Re}(s) > 1\}.$$

Having established the basic objects of study, we now introduce the key convergence concepts that will be essential for analyzing the series with functional coefficients we develop in the next section.

Definition 5 (Pointwise Convergence of a Sequence of Functions). The sequence of functions $\left\{F_n(s, z)\right\}_{n=1}^{\infty}$ converges *pointwise* to a function $F(s, z)$ on a set $E \subseteq \mathbb{C} \times D$ if for each fixed point $(s_0, z_0) \in E$, we have

$$\lim_{n \rightarrow \infty} F_n(s_0, z_0) = F(s_0, z_0).$$

That is, for each $(s_0, z_0) \in E$ and for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$, where $N = N(\varepsilon; s_0, z_0)$, such that

$$|F_n(s_0, z_0) - F(s_0, z_0)| < \varepsilon$$

for all $n \geq N$.

Specifically, let

$$F_n(s, z) = \frac{g_n(z)}{n^s},$$

then the sequence converges pointwise to $F(s, z)$ on E if

$$\lim_{n \rightarrow \infty} \frac{g_n(z_0)}{n^{s_0}} = F(s_0, z_0)$$

for each $(s_0, z_0) \in E$.

Definition 6 (Pointwise Convergence of a Series of Functions). We say that the series

$$\sum_{n=1}^{\infty} F_n(s, z)$$

converges *pointwise* on a set $E \subseteq \mathbb{C} \times D$ if for each fixed point $(s_0, z_0) \in E$, the numerical series

$$\sum_{n=1}^{\infty} F_n(s_0, z_0)$$

converges in $\mathbb{C}$. Equivalently, for each $(s_0, z_0) \in E$, the sequence of partial sums

$$H_N(s_0, z_0) = \sum_{n=1}^N F_n(s_0, z_0)$$

converges to a limit $H(s_0, z_0) \in \mathbb{C}$ as $N \rightarrow \infty$.

While pointwise convergence examines convergence at each individual point, a stronger notion of convergence requires the rate of convergence to be uniform across the entire domain.

Definition 7 (Uniform Convergence of a Sequence of Functions). The sequence of functions $\{F_n(s, z)\}_{n=1}^{\infty}$ converges *uniformly* to a function $F(s, z)$ on a set $E \subseteq \mathbb{C} \times D$ if for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$, which depends only on ε and not on the point (s, z) , such that

$$|F_n(s, z) - F(s, z)| < \varepsilon$$

for all $n \geq N$ and for all $(s, z) \in E$. Equivalently, we can also write it as follows:

$$\lim_{n \rightarrow \infty} \sup_{(s, z) \in E} |F_n(s, z) - F(s, z)| = 0.$$

Definition 8 (Uniform Convergence of a Series of Functions). The series $\sum_{n=1}^{\infty} F_n(s, z)$ converges *uniformly* to a function $h(s, z)$ on a set $E \subseteq \mathbb{C} \times D$ if the sequence of partial sums

$$H_N(s, z) = \sum_{n=1}^N F_n(s, z)$$

converges uniformly to $H(s, z)$ on E . That is, for every $\varepsilon > 0$, there exists $N_0 \in \mathbb{N}$, which depends only on ε and not on the point (s, z) , such that

$$\left| H_N(s, z) - H(s, z) \right| = \left| \sum_{n=1}^N F_n(s, z) - H(s, z) \right| = \left| \sum_{n=N+1}^{\infty} F_n(s, z) \right| < \varepsilon$$

for all $N \geq N_0$ and for all $(s, z) \in E$, where the final series expression uses the equivalent remainder term.

Definition 9 (Locally Uniform Convergence). A series $\sum_{n=1}^{\infty} F_n(s, x)$ is said to converge *locally uniformly* (or *uniformly on compact subsets*) on a set E if for every compact subset $K \subseteq E$, the series converges uniformly on K .

Remark 3. The above definitions are natural extensions of the classical theory of pointwise and uniform convergence for functions of a single complex variable to the setting of functions of two complex variables. For foundational theory in one complex variable, see Ahlfors (2021) and Rudin (1987). For the theory in several complex variables, see Krantz (2001) and Shabat (1992).

Unless otherwise stated, we henceforth restrict to real-valued z (i.e., $\text{Im}(z) = 0$), which we denote as x , to utilize properties of the hyperbolic secant function.

3 Sequence and Series Convergences

We generalize the function considered in our previous study by taking a product of exponential and hyperbolic secant functions (Karjanto 2021). Let

$$f_n(s, x) = \frac{1}{n^{s-1}} e^{-x} \operatorname{sech}^{n+1} x + \left(\frac{2-n}{n^s} \right) e^{-2x} \operatorname{sech}^n x, \quad n \in \mathbb{N}, \quad (3)$$

be a sequence of functions defined on the domain

$$E = \{(s, x) : \operatorname{Re}(s) > 1, x \geq 0\}. \quad (4)$$

In this chapter, we consider $\alpha = 1$ instead of $\alpha > 0$, and for a special case $s = 1$,

$$\lim_{s \rightarrow 1} f_n(s, x)$$

reduces to the function considered in Karjanto (2021). The convergence properties of the sequence $\{f_n(s, x)\}$ (3) and the series $\sum_{n=1}^{\infty} f_n(s, x)$ form the main subjects of this section.

Although our series $\sum_{n=1}^{\infty} f_n(s, x)$ is a Dirichlet series with functional coefficients rather than a standard Dirichlet series with multiplicative coefficients, the convergence analysis we develop shares conceptual similarities with the study of convergence abscissas for classical Dirichlet series (Brevig and Heap 2016) and spaces of translated Dirichlet series (Fernández Vidal et al. 2021). We establish both pointwise and uniform convergence properties, characterizing the domains where different modes of convergence hold.

3.1 Sequence Convergences

We have the following lemmas.

Lemma 1. *The sequence $\{f_n(s, x)\}_{n=1}^{\infty}$ defined in (3) converges pointwise to the zero function on E (4).*

Proof. We examine

$$\lim_{n \rightarrow \infty} f_n(s_0, x_0) = \lim_{n \rightarrow \infty} \frac{1}{n^{s_0-1}} e^{-x_0} \operatorname{sech}^{n+1} x_0 + \left(\frac{2-n}{n^{s_0}} \right) e^{-2x_0} \operatorname{sech}^n x_0$$

for each fixed point $(s_0, x_0) \in E$. Let us consider two cases.

Case 1: $x_0 = 0$.

From $e^{-x_0} = 1 = e^{-2x_0}$ and $\operatorname{sech} x_0 = 1$, it follows that

$$f_n(s_0, 0) = \frac{1}{n^{s_0-1}} \cdot 1 \cdot 1^{n+1} + \left(\frac{2-n}{n^{s_0}} \right) \cdot 1 \cdot 1^n = \frac{1}{n^{s_0-1}} + \frac{2}{n^{s_0}} - \frac{1}{n^{s_0-1}} = \frac{2}{n^{s_0}}.$$

Because $\operatorname{Re}(s_0) > 1$,

$$\lim_{n \rightarrow \infty} \frac{2}{n^{s_0}} = 0.$$

Case 2: $x_0 > 0$.

For $x_0 > 0$, we have $0 < \operatorname{sech}(x_0) < 1$. Let us denote $r_0 = \operatorname{sech}(x_0)$, where $0 < r_0 < 1$; then by factoring out the terms e^{-x_0} and r_0^n , we obtain

$$f_n(s_0, x_0) = e^{-x_0} r_0^n \left[\frac{r_0}{n^{s_0-1}} + \frac{(2-n)e^{-x_0}}{n^{s_0}} \right].$$

Now, as $n \rightarrow \infty$, $r_0^n \rightarrow 0$ exponentially. For sufficiently large n , we have

$$\begin{aligned} \left| \frac{r_0}{n^{s_0-1}} + \frac{(2-n)e^{-x_0}}{n^{s_0}} \right| &\leq \frac{r_0}{n^{\operatorname{Re}(s_0)-1}} + \frac{n \cdot e^{-x_0}}{n^{\operatorname{Re}(s_0)}} \\ &= \frac{r_0}{n^{\operatorname{Re}(s_0)-1}} + \frac{e^{-x_0}}{n^{\operatorname{Re}(s_0)-1}} = \frac{r_0 + e^{-x_0}}{n^{\operatorname{Re}(s_0)-1}}. \end{aligned}$$

Because $\operatorname{Re}(s_0) > 1$, this expression is $O(n^{-(\operatorname{Re}(s_0)-1)})$ with $\operatorname{Re}(s_0) - 1 > 0$, and the product with the exponential decay of r_0^n tends to zero, that is,

$$\lim_{n \rightarrow \infty} f_n(s_0, x_0) = 0.$$

This completes the proof.

We now demonstrate that the sequence converges uniformly on the entire domain E .

Lemma 2. *The sequence $\left\{ f_n(s, x) \right\}_{n=1}^{\infty}$ converges uniformly to the zero function on E (4).*

Proof. We need to verify that

$$\lim_{n \rightarrow \infty} \sup_{(s,x) \in E} |f_n(s, x) - 0| = 0,$$

or equivalently, whether for every $\varepsilon > 0$, there exists N that is independent of (s, x) such that $|f_n(s, x)| < \varepsilon$ for all $n \geq N$ and all $(s, x) \in E$.

Case 1: At $x = 0$, we have

$$|f_n(s, 0)| = \frac{2}{|n^s|} = \frac{2}{n^{\operatorname{Re}(s)}}.$$

Because $n^{\operatorname{Re}(s)}$ is decreasing as $\operatorname{Re}(s)$ decreases, taking the supremum over s with $\operatorname{Re}(s) > 1$ is achieved as $\operatorname{Re}(s) \rightarrow 1^+$, that is,

$$\sup_{\operatorname{Re}(s) > 1} |f_n(s, 0)| = \sup_{\operatorname{Re}(s) > 1} \frac{2}{n^{\operatorname{Re}(s)}} = \lim_{\operatorname{Re}(s) \rightarrow 1^+} \frac{2}{n^{\operatorname{Re}(s)}} = \frac{2}{n}.$$

Thus,

$$\lim_{n \rightarrow \infty} \sup_{(s, x) \in E} |f_n(s, 0)| = \lim_{n \rightarrow \infty} \sup_{\operatorname{Re}(s) > 1} |f_n(s, 0)| = \lim_{n \rightarrow \infty} \frac{2}{n} = 0.$$

Case 2: For $x > 0$, we bound using the triangle inequality:

$$\begin{aligned} |f_n(s, x)| &\leq \left| \frac{1}{n^{s-1}} e^{-x} \operatorname{sech}^{n+1} x \right| + \left| \frac{2-n}{n^s} e^{-2x} \operatorname{sech}^n x \right| \\ &\leq \frac{1}{n^{\operatorname{Re}(s-1)}} e^{-x} \operatorname{sech}^{n+1} x + \frac{|2-n|}{n^{\operatorname{Re}(s)}} e^{-2x} \operatorname{sech}^n x. \end{aligned}$$

For $n \geq 3$, we have $|2-n| = n-2$, and because $\operatorname{Re}(s-1) > 0$ and $\operatorname{Re}(s) > 1$, we have

$$\frac{1}{n^{\operatorname{Re}(s-1)}} \leq 1 \quad \text{and} \quad \frac{1}{n^{\operatorname{Re}(s)}} \leq \frac{1}{n}.$$

It follows that for $n \geq 3$ and $x > 0$,

$$|f_n(s, x)| \leq e^{-x} \operatorname{sech}^{n+1} x + \left(1 - \frac{2}{n}\right) e^{-2x} \operatorname{sech}^n x \leq 2 \operatorname{sech}^n x,$$

where we used $e^{-x} < 1$, $\operatorname{sech} x < 1$, and $1 - \frac{2}{n} < 1$ for $n \geq 3$. Now, for any fixed $x > 0$, we have $0 < \operatorname{sech} x < 1$, so $\operatorname{sech}^n x \rightarrow 0$ exponentially as $n \rightarrow \infty$. To establish uniform convergence over all $x > 0$, we observe that

$$\sup_{x > 0} |f_n(s, x)| \leq \sup_{x > 0} 2 \operatorname{sech}^n x = 2 \lim_{x \rightarrow 0^+} \operatorname{sech}^n x = 2.$$

However, this bound does not decay with n . The key observation is that the supremum over E combines both cases: At $x = 0$, we obtain $\frac{2}{n}$ (from Case 1), while for any fixed $x > 0$, exponential decay ensures the contribution becomes

negligible for large n . More precisely, for any $\varepsilon > 0$, choose $\delta > 0$ small enough and N large enough such that

- For $x \geq \delta$: $2 \operatorname{sech}^n \delta < \frac{\varepsilon}{2}$ for all $n \geq N$.
- For $x = 0$: $\frac{2}{n} < \frac{\varepsilon}{2}$ for all $n \geq N$.

For $0 < x < \delta$, the behavior interpolates continuously between these two regimes. Since $|f_n(s, x)|$ is continuous in x and the worst case occurs at the boundary $x = 0$ (giving $\frac{2}{n}$), we conclude

$$\sup_{(s,x) \in E} |f_n(s, x)| = \frac{2}{n}.$$

Therefore,

$$\lim_{n \rightarrow \infty} \sup_{(s,x) \in E} |f_n(s, x)| = \lim_{n \rightarrow \infty} \frac{2}{n} = 0,$$

establishing uniform convergence.

This completes the proof.

However, when we restrict the set E , the sequence $\{f_n(s, x)\}$ converges uniformly at faster rates than on the full set E . The following proposition characterizes these improved convergence rates on two natural restricted sets.

Proposition 1 (Uniform Convergence of $\{f_n(s, x)\}$ on Restricted Sets). *Let*

$$E_\delta = \{(s, x) : \operatorname{Re}(s) \geq 1 + \delta, x \geq 0\}, \quad \delta > 0,$$

and

$$E_\varepsilon = \{(s, x) : \operatorname{Re}(s) > 1, x \geq \varepsilon\}, \quad \varepsilon > 0.$$

Then the sequence $\{f_n(s, x)\}_{n=1}^\infty$ converges uniformly on both sets:

1. On E_δ , the sequence converges uniformly with polynomial rate $O(1/n^{1+\delta})$.
2. On E_ε , the sequence converges uniformly with exponential rate $O(r_\varepsilon^n)$, where $r_\varepsilon = \operatorname{sech} \varepsilon \in (0, 1)$.

In particular, the exponential rate on E_ε is faster than the polynomial rate on E_δ .

Proof. We prove the uniform convergence for each restricted set separately.

1. Because $\operatorname{Re}(s) \geq 1 + \delta$ on E_δ , at $x = 0$ we have

$$|f_n(s, 0)| = \frac{2}{n^{\operatorname{Re}(s)}} \leq \frac{2}{n^{1+\delta}}.$$

For $x > 0$, following the same line of argument as in Lemma 2, we use the triangle inequality and the constraint $\operatorname{Re}(s) \geq 1 + \delta$ to obtain for $n \geq 3$:

$$\begin{aligned} |f_n(s, x)| &\leq \frac{1}{n^{\operatorname{Re}(s-1)}} e^{-x} \operatorname{sech}^{n+1} x + \frac{n-2}{n^{\operatorname{Re}(s)}} e^{-2x} \operatorname{sech}^n x \\ &\leq \frac{1}{n^\delta} e^{-x} \operatorname{sech}^{n+1} x + \frac{n-2}{n^{1+\delta}} e^{-2x} \operatorname{sech}^n x \\ &\leq \frac{1}{n^\delta} \operatorname{sech}^n x \left[e^{-x} \operatorname{sech} x + e^{-2x} \right] \\ &\leq \frac{2}{n^\delta} \operatorname{sech}^n x, \end{aligned}$$

where we used $e^{-x} < 1$, and $e^{-2x} < 1$ for $x > 0$. Now, for any fixed $x > 0$, since $0 < \operatorname{sech} x < 1$, the term $\operatorname{sech}^n x$ decays exponentially. This exponential decay is much faster than the polynomial growth n^δ :

$$\lim_{n \rightarrow \infty} n^\delta \cdot \operatorname{sech}^n x = 0.$$

Therefore, for any fixed $x > 0$, there exists $N_0(x, \delta)$ such that for all $n \geq N_0$

$$\frac{2}{n^\delta} \operatorname{sech}^n x \leq \frac{2}{n^{1+\delta}}.$$

Combining both cases, the supremum over E_δ is dominated by the boundary behavior at $x = 0$:

$$\sup_{(s,x) \in E_\delta} |f_n(s, x)| \leq \frac{C}{n^{1+\delta}},$$

for some constant C independent of n . Thus, the sequence converges uniformly on E_δ with polynomial rate $O(1/n^{1+\delta})$.

2. Let $r_\varepsilon = \operatorname{sech} \varepsilon$, then $0 < r_\varepsilon < 1$. From our earlier analysis in Lemma 2, we have

$$|f_n(s, x)| \leq \operatorname{sech}^n x \left[e^{-x} \operatorname{sech} x + e^{-2x} \left(1 - \frac{2}{n} \right) \right].$$

For $x \geq \varepsilon > 0$, we have $\operatorname{sech} x \leq \operatorname{sech} \varepsilon$, and thus $\operatorname{sech}^n x \leq r_\varepsilon^n$, and the term in the brackets bounds as

$$e^{-x} \operatorname{sech} x + e^{-2x} \left(1 - \frac{2}{n} \right) \leq e^{-\varepsilon} \operatorname{sech} \varepsilon + e^{-2\varepsilon} \leq e^{-\varepsilon} (1 + e^{-\varepsilon}) \leq 2e^{-\varepsilon},$$

because $1 - 2/n \leq 1$ for $n \geq 2$. Taking the limit and supremum, we obtain

$$\lim_{n \rightarrow \infty} \sup_{(s,x) \in E_\varepsilon} |f_n(s, x)| \leq \lim_{n \rightarrow \infty} 2e^{-\varepsilon} r_\varepsilon^n = 2e^{-\varepsilon} \cdot 0 = 0.$$

Hence, the sequence converges uniformly on E_ε with exponential convergence rate $O(r_\varepsilon^n)$, where $r_\varepsilon = \text{sech } \varepsilon \in (0, 1)$.

This completes the proof.

With the convergence properties of the sequence $\{f_n(s, x)\}$ fully characterized, we now turn our attention to the convergence of the series $\sum_{n=1}^{\infty} f_n(s, x)$ itself.

3.2 Series Convergences

Theorem 2 (Pointwise Convergence of the Series $\sum_{n=1}^{\infty} f_n(s, x)$). *Let $f_n(s, x)$ be as defined in (3) on the domain E (4); then the series*

$$\sum_{n=1}^{\infty} f_n(s, x)$$

converges pointwise on E to a function $h(s, x)$ given by

$$h(s, x) = \begin{cases} 2\zeta(s), & x = 0, \\ e^{-x} (\text{sech } x - e^{-x}) \text{Li}_{s-1}(\text{sech } x) + 2e^{-2x} \text{Li}_s(\text{sech } x), & x > 0, \end{cases} \quad (5)$$

where $\zeta(s)$ is the Riemann zeta function (1) and $\text{Li}_s(z)$ is the polylogarithm function (2).

Proof. We split the proof into two cases.

Case 1: At $x = 0$, we know from our earlier analysis that

$$S_N(s, 0) = \sum_{n=1}^N f_n(s, 0) = \sum_{n=1}^N \frac{2}{n^s}.$$

Because $\text{Re}(s) > 1$, this is a Dirichlet partial sum with all coefficients equal to 2. The associated Dirichlet series converges to $2\zeta(s)$ for $N \rightarrow \infty$.

Case 2: For $x > 0$, we derive an upper bound for the Dirichlet series with functional coefficients and calculate an exact expression using polylogarithm functions.

From Lemma 2, we obtain the following partial sum by applying the triangle inequality:

$$|S_N(s, x)| \leq \sum_{n=1}^N |f_n(s, x)| \leq e^{-x} \underbrace{\sum_{n=1}^N \frac{\operatorname{sech}^{n+1} x}{n^{\operatorname{Re}(s-1)}}}_{P_1(s, x)} + e^{-2x} \underbrace{\sum_{n=1}^N \frac{|2-n|}{n^{\operatorname{Re}(s)}} \operatorname{sech}^n x}_{P_2(s, x)}. \quad (6)$$

We carefully bound the right-hand side of inequality (6) by estimating each of its partial sums. Using $n^{\operatorname{Re}(s-1)} \geq 1$ for all $n \geq 1$, the first partial sum $P_1(s, x)$ from the right-hand side of inequality (6) can be bounded by

$$P_1(s, x) = \sum_{n=1}^N \frac{\operatorname{sech}^{n+1} x}{n^{\operatorname{Re}(s-1)}} \leq \sum_{n=1}^N \operatorname{sech}^{n+1} x = \operatorname{sech}^2 x \frac{1 - \operatorname{sech}^N x}{1 - \operatorname{sech} x}.$$

Similarly, by splitting the summation based on the absolute value sign, using $n - 2 \leq n$ for all $n \geq 3$, and $n^{\operatorname{Re}(s)-1} \geq 1$ for all $n \geq 1$, the second partial sum $P_2(s, x)$ from the right-hand side of inequality (6) can be written as follows:

$$\begin{aligned} P_2(s, x) &= \sum_{n=1}^N \frac{|2-n|}{n^{\operatorname{Re}(s)}} \operatorname{sech}^n x = \sum_{n=1}^2 \frac{2-n}{n^{\operatorname{Re}(s)}} \operatorname{sech}^n x + \sum_{n=3}^N \frac{n-2}{n^{\operatorname{Re}(s)}} \operatorname{sech}^n x \\ P_2(s, x) &\leq \operatorname{sech} x + \sum_{n=3}^N \frac{n}{n^{\operatorname{Re}(s)}} \operatorname{sech}^n x \\ &\leq \sum_{n=1}^N \frac{1}{n^{\operatorname{Re}(s)-1}} \operatorname{sech}^n x \\ &\leq \sum_{n=1}^N \operatorname{sech}^n x = \operatorname{sech} x \frac{1 - \operatorname{sech}^N x}{1 - \operatorname{sech} x}. \end{aligned}$$

Combining both bounds and taking $N \rightarrow \infty$, as well as noting that $\operatorname{sech}^N x \rightarrow 0$ because $0 < \operatorname{sech} x < 1$ for $x > 0$, it follows that

$$\begin{aligned} |h(s, x)| &= \left| \lim_{N \rightarrow \infty} S_N(s, x) \right| \leq \lim_{N \rightarrow \infty} |S_N(s, x)| \\ &\leq \lim_{N \rightarrow \infty} \left(e^{-x} \operatorname{sech}^2 x + e^{-2x} \operatorname{sech} x \right) \frac{1 - \operatorname{sech}^N x}{1 - \operatorname{sech} x} \\ &= e^{-x} \operatorname{sech} x \left(\frac{e^{-x} + \operatorname{sech} x}{1 - \operatorname{sech} x} \right). \end{aligned}$$

Because each term $|f_n(s, x)|$ decays exponentially due to the factor $\operatorname{sech}^n x$ with $0 < \operatorname{sech} x < 1$, this bound establishes that the partial sums $S_N(s, x)$ remain bounded as $N \rightarrow \infty$, which confirms that the series $\sum_{n=1}^{\infty} f_n(s, x)$ converges absolutely for each fixed $(s, x) \in E$ with $x > 0$, establishing the existence and finiteness of the limit $h(s, x)$.

Having established pointwise convergence, we now determine the explicit form of $h(s, x)$ by recognizing the series as a linear combination of polylogarithm functions. The absolute convergence established above allows us to rearrange terms and factor out constants freely. We split the terms in the original Dirichlet series:

$$\begin{aligned} h(s, x) &= e^{-x} \operatorname{sech} x \sum_{n=1}^{\infty} \frac{\operatorname{sech}^n x}{n^{s-1}} + 2e^{-2x} \sum_{n=1}^{\infty} \frac{\operatorname{sech}^n x}{n^s} - e^{-2x} \sum_{n=1}^{\infty} \frac{\operatorname{sech}^n x}{n^{s-1}} \\ &= e^{-x} \operatorname{sech} x \operatorname{Li}_{s-1}(\operatorname{sech} x) + 2e^{-2x} \operatorname{Li}_s(\operatorname{sech} x) \\ &\quad - e^{-2x} \operatorname{Li}_{s-1}(\operatorname{sech} x) \\ &= e^{-x} (\operatorname{sech} x - e^{-x}) \operatorname{Li}_{s-1}(\operatorname{sech} x) + 2e^{-2x} \operatorname{Li}_s(\operatorname{sech} x), \end{aligned}$$

where $\operatorname{Li}_s(z)$ is the polylogarithm function (2). Because $|\operatorname{sech} x| < 1$, both $\operatorname{Li}_{s-1}(\operatorname{sech} x)$ and $\operatorname{Li}_s(\operatorname{sech} x)$ converge absolutely for all $s \in \mathbb{C}$ with $\operatorname{Re}(s) > 1$.

Therefore, the series $\sum_{n=1}^{\infty} f_n(s, x)$ converges pointwise for all $(s, x) \in E$. This completes the proof.

While Theorem 2 establishes pointwise convergence of the series on E , a natural question arises: Does this series also converge uniformly on E ? The following theorem shows that it does not.

Theorem 3 (Nonuniform Convergence on E). *The series $\sum_{n=1}^{\infty} f_n(s, x)$ does not converge uniformly on the domain E (4).*

Proof. We prove nonuniform convergence by showing that the supremum of the series' remainder (tail) cannot be made uniformly small over all of E . For uniform convergence, we would need

$$\lim_{N \rightarrow \infty} \sup_{(s, x) \in E} \left| \sum_{n=N+1}^{\infty} f_n(s, x) \right| = 0.$$

We prove this fails by examining the behavior at $x = 0$ with real values $s = \sigma \in \mathbb{R}$ where $\sigma > 1$. At $x = 0$, the tail of the series is

$$\left| \sum_{n=N+1}^{\infty} f_n(\sigma, 0) \right| = \left| \sum_{n=N+1}^{\infty} \frac{2}{n^\sigma} \right| = 2 \sum_{n=N+1}^{\infty} \frac{1}{n^\sigma}.$$

To obtain a lower bound on this sum, we use the integral test. Observe that for the decreasing function $f(t) = 1/t^\sigma$, the sum can be approximated by rectangles, and the left end point rectangles give an overestimate of the integral. Therefore,

$$\sum_{n=N+1}^{\infty} \frac{1}{n^\sigma} \geq \int_{N+1}^{\infty} \frac{1}{t^\sigma} dt = \lim_{b \rightarrow \infty} \left[\frac{t^{1-\sigma}}{1-\sigma} \right]_{N+1}^b = \frac{1}{(\sigma-1)(N+1)^{\sigma-1}}.$$

It follows that

$$\left| \sum_{n=N+1}^{\infty} f_n(\sigma, 0) \right| \geq \frac{2}{(\sigma-1)(N+1)^{\sigma-1}}.$$

Let $E_0 = \{(\sigma, 0) : \sigma \in \mathbb{R}, \sigma > 1\}$; then it is obvious that the set of points with real $s > 1$ and $x = 0$ is a subset of all points in E (4), i.e., $E_0 \subseteq E$. Therefore, taking the supremum over $(s, x) \in E$ and noting that the supremum over $x \geq 0$ includes $x = 0$,

$$\begin{aligned} \sup_{(s,x) \in E} \left| \sum_{n=N+1}^{\infty} f_n(s, x) \right| &\geq \sup_{(\sigma,0) \in E_0} \left| \sum_{n=N+1}^{\infty} f_n(\sigma, 0) \right| = \sup_{\sigma > 1} \left| \sum_{n=N+1}^{\infty} f_n(\sigma, 0) \right| \\ &\geq \sup_{\sigma > 1} \frac{2}{(\sigma-1)(N+1)^{\sigma-1}}, \end{aligned}$$

where the last inequality follows from the integral test bound. As $\sigma \rightarrow 1^+$,

$$\lim_{\sigma \rightarrow 1^+} \frac{1}{(\sigma-1)(N+1)^{\sigma-1}} = +\infty,$$

because $(N+1)^{\sigma-1} \rightarrow (N+1)^0 = 1$, whereas $(\sigma-1) \rightarrow 0^+$. Therefore, for any fixed N ,

$$\sup_{(s,x) \in E} \left| \sum_{n=N+1}^{\infty} f_n(s, x) \right| = +\infty.$$

This implies

$$\lim_{N \rightarrow \infty} \sup_{(s,x) \in E} \left| \sum_{n=N+1}^{\infty} f_n(s, x) \right| = +\infty \neq 0,$$

which shows that the series does not converge uniformly on E . The failure of uniform convergence occurs because as $\operatorname{Re}(s) \rightarrow 1^+$, i.e., approaching the boundary of the convergence region, the tail of the series at $x = 0$ grows without bound, regardless of how large N is chosen. This completes the proof.

The following remark clarifies the nonuniformity mechanism by distinguishing the roles of s and x in determining convergence.

Remark 4. The failure of uniform convergence is due to the lack of a sufficient decay rate at the $x = 0$ boundary point as the s variable approaches the critical line $\operatorname{Re}(s) = 1$. This phenomenon is analogous to the classical behavior of Dirichlet series near their abscissa of convergence. For Dirichlet series with multiplicative coefficients, Brevig and Heap (2016) established that the abscissas of uniform and absolute convergence coincide, a property that does not generally hold for arbitrary Dirichlet series. While our series has functional coefficients and thus falls outside their framework, the restriction to subsets E_δ and E_ε in Theorem 4 can be viewed as an analogous strategy for achieving uniform convergence away from critical boundaries. The proof of Theorem 3 clearly shows that the failure of uniform convergence is entirely localized to the boundary of the abscissa of convergence, where $\operatorname{Re}(s) \rightarrow 1^+$.

- The role of s :
The nonuniformity is caused by the tail of the series at $x = 0$ reducing to the tail of the series defining $2\zeta(s)$ (1), that is,

$$\sum_{n=N+1}^{\infty} \frac{2}{n^\sigma}.$$

As $\sigma \rightarrow 1^+$, this remainder term diverges to infinity, violating the condition for uniform convergence. This is a classic behavior for Dirichlet series at the boundary of their convergence half plane.

- The role of x :
For any fixed $x > 0$, the term $\operatorname{sech}^n x$ introduces exponential decay in n , which is fast enough to ensure uniform convergence, as proven on E_ε in Theorem 4. The variable x therefore does not contribute to the failure of uniform convergence. Instead, it provides a stabilizing geometric factor that dominates the polynomial decay, except at the single point $x = 0$.

Although the series does not converge uniformly on the entire set E (4), it does converge uniformly on appropriately restricted subsets, as the following theorem demonstrates.

Theorem 4 (Uniform Convergence on Restricted Areas). *For any $\delta > 0$ and $\varepsilon > 0$, let E_δ and E_ε be two sets introduced in Proposition 1; then the series*

$$\sum_{n=1}^{\infty} f_n(s, x) = \sum_{n=1}^{\infty} \left(\frac{1}{n^{s-1}} e^{-x} \operatorname{sech}^{n+1} x + \frac{2-n}{n^s} e^{-2x} \operatorname{sech}^n x \right)$$

converges uniformly on both sets.

Proof. We use the Weierstrass M-test in both cases.

1. Uniform convergence on E_δ .

We need to find a convergent series of constants $\sum_{n=1}^{\infty} M_{n,\delta}$ such that $|f_n(s, x)| \leq M_{n,\delta}$ for all $(s, x) \in E_\delta$ and all $n \in \mathbb{N}$. From our earlier analysis in Proposition 1, we established that for $(s, x) \in E_\delta$ with $\operatorname{Re}(s) \geq 1 + \delta$:

- At $x = 0$,

$$|f_n(s, 0)| = \frac{2}{n^{\operatorname{Re}(s)}} \leq \frac{2}{n^{1+\delta}}.$$

- For $x > 0$ and $n \geq 3$,

$$|f_n(s, x)| \leq \frac{2}{n^\delta} \operatorname{sech}^n x.$$

For any fixed $x > 0$, the exponential decay $\operatorname{sech}^n x \rightarrow 0$ (because $0 < \operatorname{sech} x < 1$) dominates the polynomial factor n^δ , ensuring that

$$\lim_{n \rightarrow \infty} n^\delta \cdot \operatorname{sech}^n x = 0.$$

Therefore, for sufficiently large n , we have $\frac{2}{n^\delta} \operatorname{sech}^n x \leq \frac{2}{n^{1+\delta}}$, and the supremum over all $(s, x) \in E_\delta$ is dominated by the boundary case at $x = 0$. Thus, for all $(s, x) \in E_\delta$ and all $n \in \mathbb{N}$, we can set

$$M_{n,\delta} := \frac{C}{n^{1+\delta}},$$

where $C \geq 2$ is a constant accounting for the initial terms. Because $\delta > 0$, the series

$$\sum_{n=1}^{\infty} M_{n,\delta} = C \sum_{n=1}^{\infty} \frac{1}{n^{1+\delta}} = C\zeta(1+\delta) < \infty$$

converges. Hence, by the Weierstrass M-test, the series $\sum_{n=1}^{\infty} f_n(s, x)$ converges uniformly on E_{δ} .

2. Uniform convergence on E_{ε} .

Similarly, we also need to find a convergent series of constants $\sum_{n=1}^{\infty} M_{n,\varepsilon}$ such that $|f_n(s, x)| \leq M_{n,\varepsilon}$ for all $(s, x) \in E_{\varepsilon}$ and all $n \in \mathbb{N}$. From our earlier analysis in Proposition 1, for sufficiently large n and for $(s, x) \in E_{\varepsilon}$, the modulus of the function sequence terms can be bounded by

$$|f_n(s, x)| \leq M_{n,\varepsilon} := 2e^{-\varepsilon} \operatorname{sech}^n \varepsilon.$$

Because $0 < \operatorname{sech} \varepsilon < 1$, it follows that the geometric series

$$\sum_{n=1}^{\infty} 2e^{-\varepsilon} \operatorname{sech}^n \varepsilon = \frac{2e^{-\varepsilon} \operatorname{sech} \varepsilon}{1 - \operatorname{sech} \varepsilon} < \infty$$

converges. Thus, by the Weierstrass M-test, the series $\sum_{n=1}^{\infty} f_n(s, x)$ also converges uniformly on E_{ε} .

This completes the proof.

Based on the results of Theorems 3 and 4, the following remark summarizes the dual nature of the series' convergence and its classification as locally uniform.

Remark 5. The following two points consolidate the convergence analysis of the series, clarifying the reasons for the failure of uniform convergence on E and providing the precise classification of its global behavior.

- Although Theorem 3 proves that the series does not converge uniformly on the entire set E , it does converge uniformly when the domain is appropriately restricted, that is, either by requiring $\operatorname{Re}(s)$ to be bounded away from the critical value 1 (giving rise to E_{δ}) or by restricting x to be bounded away from 0 (giving rise to E_{ε}), as shown in Theorem 4. Consequently, the convergence behavior across the full domain E is classified as *locally uniform convergence* (or *uniform convergence on compact subsets*), meaning the series converges uniformly on every compact subset K of E . This is a standard mode of convergence for series of analytic functions (Ahlfors 2021; Krantz 2001). See also Definition 9.
- It is further noteworthy that the convergence on E_{ε} is exponentially fast (rate $O(r_{\varepsilon}^n)$), owing to the geometric nature of the series, whereas the convergence on E_{δ} is polynomially fast due to the p -series with $p = 1 + \delta$ (rate $O(1/n^{1+\delta})$), controlled by the boundary case $x = 0$.

As an immediate consequence of the uniform convergence on E_ε , we can explicitly identify the limit function.

Corollary 1 (Explicit Uniform Limit on E_ε). *For any $\varepsilon > 0$, the series $\sum_{n=1}^{\infty} f_n(s, x)$ converges uniformly on E_ε to the function $h(s, x)$ given in (5):*

$$h(s, x) = e^{-x} (\operatorname{sech} x - e^{-x}) \operatorname{Li}_{s-1}(\operatorname{sech} x) + 2e^{-2x} \operatorname{Li}_s(\operatorname{sech} x),$$

where $\operatorname{Li}_s(z)$ is the polylogarithm function (2).

4 Integral Representations

In this section, we develop integral representations of the sum function $h(s, x)$ (5) by integrating term by term. The uniform convergence established in Theorem 4 on E_ε provides the necessary foundation for interchanging integration and summation. Our main result will express integrals of $h(s, x)$ explicitly in terms of the polylogarithm function (2). We begin with the following general result.

4.1 Interchange of Integration and Summation

We begin with a general result that justifies the interchange of integration and summation for series of functions.

Theorem 5 (Interchange of Integration and Summation). *Let $\varepsilon > 0$ and fix $s \in \mathbb{C}$ with $\operatorname{Re}(s) > 1$. Consider the series*

$$h(s, x) = \sum_{n=1}^{\infty} f_n(s, x)$$

on the restricted domain $E_\varepsilon \subset E$ (4), where $E_\varepsilon = \{(s, x) : \operatorname{Re}(s) > 1, x \geq \varepsilon\}$. Then for any a and b with $\varepsilon \leq a < b < \infty$, we have

$$\int_a^b h(s, x) dx = \int_a^b \sum_{n=1}^{\infty} f_n(s, x) dx = \sum_{n=1}^{\infty} \int_a^b f_n(s, x) dx.$$

Moreover, for the improper integral over $[\varepsilon, \infty)$, if the following two conditions hold:

(i) *The improper integral*

$$\int_{\varepsilon}^{\infty} h(s, x) dx \text{ converges, i.e., } \lim_{b \rightarrow \infty} \int_{\varepsilon}^b h(s, x) dx \text{ exists}$$

(ii) *The series*

$$\sum_{n=1}^{\infty} \int_{\varepsilon}^{\infty} f_n(s, x) dx \text{ converges,}$$

then

$$\int_{\varepsilon}^{\infty} h(s, x) dx = \sum_{n=1}^{\infty} \int_{\varepsilon}^{\infty} f_n(s, x) dx.$$

Proof. We prove the parts for a finite and infinite intervals separately.

- For a finite interval $[a, b]$:

From Theorem 4, we know that the series $\sum_{n=1}^{\infty} f_n(s, x)$ converges uniformly to $h(s, x)$ on E_{ε} . For any fixed s with $\operatorname{Re}(s) > 1$, the uniform convergence on E_{ε} implies uniform convergence on the compact interval $[a, b] \subset [\varepsilon, \infty)$ as a function of x alone. Using the uniform convergence theorem for Riemann integrals, if a sequence of functions converges uniformly on $[a, b]$ and each f_n is Riemann integrable, then the limit can be interchanged with the integral:

$$\lim_{N \rightarrow \infty} \int_a^b S_N(s, x) dx = \int_a^b \lim_{N \rightarrow \infty} S_N(s, x) dx,$$

where $S_N(s, x) = \sum_{n=1}^N f_n(s, x)$ is the N -th partial sum. Because $\lim_{N \rightarrow \infty} S_N(s, x) = h(s, x)$, we have

$$\int_a^b h(s, x) dx = \lim_{N \rightarrow \infty} \int_a^b S_N(s, x) dx = \lim_{N \rightarrow \infty} \int_a^b \sum_{n=1}^N f_n(s, x) dx.$$

By linearity of integration, we obtain

$$\int_a^b \sum_{n=1}^N f_n(s, x) dx = \sum_{n=1}^N \int_a^b f_n(s, x) dx.$$

The passage to the infinite series is justified by the uniform convergence on $[a, b]$, which ensures that

$$\begin{aligned} & \lim_{N \rightarrow \infty} \left| \int_a^b S_N(s, x) dx - \int_a^b h(s, x) dx \right| \\ &= \lim_{N \rightarrow \infty} \left| \int_a^b [S_N(s, x) - h(s, x)] dx \right| \\ &\leq \lim_{N \rightarrow \infty} (b - a) \sup_{x \in [a, b]} |S_N(s, x) - h(s, x)| = 0. \end{aligned}$$

Therefore,

$$\int_a^b h(s, x) dx = \lim_{N \rightarrow \infty} \sum_{n=1}^N \int_a^b f_n(s, x) dx = \sum_{n=1}^{\infty} \int_a^b f_n(s, x) dx.$$

- For an improper integral over $[\varepsilon, \infty)$:

Assume conditions (i) and (ii) hold. From the first item, for any $b > \varepsilon$, we have

$$\int_{\varepsilon}^b h(s, x) dx = \sum_{n=1}^{\infty} \int_{\varepsilon}^b f_n(s, x) dx.$$

Using condition (i), taking the limit as $b \rightarrow \infty$ on the left-hand side of the equation yields

$$\lim_{b \rightarrow \infty} \int_{\varepsilon}^b h(s, x) dx = \int_{\varepsilon}^{\infty} h(s, x) dx.$$

For the right-hand side, we interchange the limit $b \rightarrow \infty$ with the infinite sum. Observe that

$$\lim_{b \rightarrow \infty} \sum_{n=1}^{\infty} \int_{\varepsilon}^b f_n(s, x) dx = \sum_{n=1}^{\infty} \lim_{b \rightarrow \infty} \int_{\varepsilon}^b f_n(s, x) dx = \sum_{n=1}^{\infty} \int_{\varepsilon}^{\infty} f_n(s, x) dx.$$

This interchange of limit and summation is justified because (a) the series converges uniformly on each compact subinterval $[\varepsilon, b]$ by Theorem 4, which ensures the interchange is valid for each finite b , and (b) condition (ii) ensures that the series of limiting integrals converges, which allows us to pass to the limit as $b \rightarrow \infty$. Hence,

$$\int_{\varepsilon}^{\infty} h(s, x) dx = \sum_{n=1}^{\infty} \int_{\varepsilon}^{\infty} f_n(s, x) dx.$$

This completes the proof.

Remark 6 (Alternative Approach Via Weierstrass M -Test). The interchange of limit and summation in the second item of the proof of Theorem 5 can also be established using the Weierstrass M -test, which provides explicit bounds on the rate of convergence and offers a more direct verification of uniform convergence. The goal is to show that the series $\sum_{n=1}^{\infty} \int_{\varepsilon}^b f_n(s, x) dx$ converges uniformly in b , which will allow us to interchange the limit $b \rightarrow \infty$ with the series. We use the Weierstrass M -test by finding constants M_n such that

$$\left| \int_{\varepsilon}^b f_n(s, x) dx \right| \leq M_n \quad \text{for all } b > \varepsilon,$$

$$\text{and} \quad \sum_{n=1}^{\infty} M_n < \infty.$$

We claim that $M_n = C_n(1 - \tanh \varepsilon)$ works, where

$$C_n = e^{-\varepsilon} \operatorname{sech}^{n+1} \varepsilon [1 + e^{-\varepsilon} \cosh \varepsilon].$$

For $x \geq \varepsilon > 0$, we have $\operatorname{sech} x \leq \operatorname{sech} \varepsilon < 1$ and $e^{-x} \leq e^{-\varepsilon} < 1$. From the analysis in Theorem 4, for $n \geq 3$,

$$\begin{aligned} |f_n(s, x)| &\leq e^{-x} \operatorname{sech}^{n+1} x + \left(1 - \frac{2}{n}\right) e^{-2x} \operatorname{sech}^n x \\ &\leq e^{-x} \operatorname{sech}^{n+1} x + e^{-2x} \operatorname{sech}^n x \\ &= e^{-x} \operatorname{sech}^n x (\operatorname{sech} x + e^{-x}) \\ &\leq e^{-\varepsilon} \operatorname{sech}^{n+1} \varepsilon (1 + e^{-\varepsilon} \cosh \varepsilon) \operatorname{sech}^2 x \\ &= C_n \operatorname{sech}^2 x, \end{aligned}$$

where in the last inequality we used $\operatorname{sech} x + e^{-x} = \frac{2}{e^x + e^{-x}} + e^{-x} \leq 1 + e^{-\varepsilon} \cosh \varepsilon$ for $x \geq \varepsilon$. Therefore, for any $b > \varepsilon$,

$$\begin{aligned} \left| \int_{\varepsilon}^b f_n(s, x) dx \right| &\leq \int_{\varepsilon}^b |f_n(s, x)| dx \\ &\leq \int_{\varepsilon}^b C_n \operatorname{sech}^2 x dx \\ &= C_n [\tanh x]_{\varepsilon}^b \\ &= C_n (\tanh b - \tanh \varepsilon) \\ &\leq C_n (1 - \tanh \varepsilon) =: M_n, \end{aligned}$$

where we used $\tanh b < 1$ for all finite b . To verify that $\sum_{n=1}^{\infty} M_n$ converges, we compute

$$\begin{aligned} \sum_{n=1}^{\infty} M_n &= (1 - \tanh \varepsilon) e^{-\varepsilon} (1 + e^{-\varepsilon} \cosh \varepsilon) \sum_{n=1}^{\infty} \operatorname{sech}^{n+1} \varepsilon \\ &= (1 - \tanh \varepsilon) e^{-\varepsilon} (1 + e^{-\varepsilon} \cosh \varepsilon) \cdot \frac{\operatorname{sech}^2 \varepsilon}{1 - \operatorname{sech} \varepsilon} < \infty, \end{aligned}$$

where the geometric series converges because $0 < \operatorname{sech} \varepsilon < 1$ for $\varepsilon > 0$. By the Weierstrass M -test, the series $\sum_{n=1}^{\infty} \int_{\varepsilon}^b f_n(s, x) dx$ converges uniformly in b , which justifies the interchange

$$\lim_{b \rightarrow \infty} \sum_{n=1}^{\infty} \int_{\varepsilon}^b f_n(s, x) dx = \sum_{n=1}^{\infty} \lim_{b \rightarrow \infty} \int_{\varepsilon}^b f_n(s, x) dx = \sum_{n=1}^{\infty} \int_{\varepsilon}^{\infty} f_n(s, x) dx.$$

This alternative approach has the advantage of providing explicit bounds on the rate of convergence of the series of integrals.

To apply this interchange theorem effectively, we first need to compute the indefinite integral of each term of the sequence $f_n(s, x)$ (3).

Lemma 3. *For $n \in \mathbb{N}$ and $(s, x) \in E$, the indefinite integral of $f_n(s, x)$ with respect to x is given by*

$$F_n(s, x) := \int f_n(s, x) dx = -\frac{e^{-2x}}{n^s} \operatorname{sech}^n x + C,$$

where C is the integration constant.

Proof. Let us integrate the function $f_n(s, x)$ (3) with respect to x :

$$\begin{aligned} \int f_n(s, x) dx &= \int \left(\frac{1}{n^{s-1}} e^{-x} \operatorname{sech}^{n+1} x + \left(\frac{2-n}{n^s} \right) e^{-2x} \operatorname{sech}^n x \right) dx \\ &= \int \left(\frac{n}{n^s} e^{-x} \operatorname{sech}^{n+1} x + \frac{2}{n^s} e^{-2x} \operatorname{sech}^n x - \frac{n}{n^s} e^{-2x} \operatorname{sech}^n x \right) dx \\ &= \int \left(\frac{2}{n^s} e^{-2x} \operatorname{sech}^n x + \frac{n}{n^s} e^{-x} \operatorname{sech}^{n+1} x \right. \\ &\quad \left. - \frac{1}{2} \frac{n}{n^s} (e^x + e^{-x}) \operatorname{sech} x e^{-2x} \operatorname{sech}^n x \right) dx \end{aligned}$$

$$\begin{aligned}
&= \int \left(\frac{2}{n^s} e^{-2x} \operatorname{sech}^n x + \frac{n}{n^s} e^{-x} \operatorname{sech}^{n+1} x - \frac{1}{2} \frac{n}{n^s} e^{-x} \operatorname{sech}^{n+1} x \right. \\
&\quad \left. - \frac{1}{2} \frac{n}{n^s} e^{-3x} \operatorname{sech}^{n+1} x \right) dx \\
&= \int \left(\frac{2}{n^s} e^{-2x} \operatorname{sech}^n x + \frac{1}{2} \frac{n}{n^s} e^{-x} \operatorname{sech}^{n+1} x \right. \\
&\quad \left. - \frac{1}{2} \frac{n}{n^s} e^{-3x} \operatorname{sech}^{n+1} x \right) dx \\
&= \int \left(\frac{2}{n^s} e^{-2x} \operatorname{sech}^n x + \frac{n}{n^s} e^{-2x} \operatorname{sech}^{n+1} x \cdot \frac{1}{2} (e^x - e^{-x}) \right) dx \\
&= \int \left(\frac{2}{n^s} e^{-2x} \operatorname{sech}^n x - \frac{n}{n^s} e^{-2x} \operatorname{sech}^{n-1} x \cdot (-\operatorname{sech} x \tanh x) \right) dx \\
&= \int \left(-\frac{1}{n^s} \frac{d}{dx} (e^{-2x}) \operatorname{sech}^n x - \frac{n}{n^s} e^{-2x} \operatorname{sech}^{n-1} x \right. \\
&\quad \left. \cdot \frac{d}{dx} (\operatorname{sech} x) \right) dx \\
&= \int \frac{d}{dx} \left(-\frac{1}{n^s} e^{-2x} \operatorname{sech}^n x \right) dx \\
\int f_n(s, x) dx &= -\frac{e^{-2x}}{n^s} \operatorname{sech}^n x + C := F_n(s, x).
\end{aligned}$$

This completes the proof.

The direct integration in the proof of Lemma 3 involves several algebraic manipulations. For verification purposes, we note that an alternative approach exists.

Remark 7. Alternatively, we can also prove the lemma by direct partial differentiation of $F_n(s, x)$ with respect to x , that is, verifying that

$$\frac{\partial}{\partial x} F_n(s, x) = f_n(s, x)$$

for all $n \in \mathbb{N}$. The computation can be performed by applying the product rule and manipulating algebraic identities involving hyperbolic functions.

Using the antiderivative $F_n(s, x)$ from Lemma 3, we can now compute the definite integral of $f_n(s, x)$ over any interval.

Corollary 2 (Definite Integral of $f_n(s, x)$). For $\varepsilon > 0$ and $b > \varepsilon$, the definite integral of $f_n(s, x)$ (3) over $[\varepsilon, b]$ is given by

$$\int_{\varepsilon}^b f_n(s, x) dx = -\frac{e^{-2b}}{n^s} \operatorname{sech}^n b + \frac{e^{-2\varepsilon}}{n^s} \operatorname{sech}^n \varepsilon.$$

Moreover, as $b \rightarrow \infty$, we denote this limiting integral by

$$\widehat{F}_n(s, \varepsilon) := \int_{\varepsilon}^{\infty} f_n(s, x) dx = \frac{e^{-2\varepsilon}}{n^s} \operatorname{sech}^n \varepsilon$$

because $e^{-2b} \operatorname{sech}^n b \rightarrow 0$ as $b \rightarrow \infty$ for $n \geq 1$.

4.2 Integral Representation and Explicit Formulas

Having computed the integrals of the individual terms $f_n(s, x)$ (3) in the previous subsection, we now establish the convergence of the series $\sum_{n=1}^{\infty} \widehat{F}_n(s, \varepsilon)$ and derive an explicit integral representation for $h(s, x)$ in terms of the polylogarithm function.

Theorem 6 (Integral representation of $h(s, x)$). For $\varepsilon > 0$ and $s \in \mathbb{C}$ with $\operatorname{Re}(s) > 1$, the following hold:

1. The series

$$\sum_{n=1}^{\infty} \int_{\varepsilon}^{\infty} f_n(s, x) dx = \sum_{n=1}^{\infty} \frac{e^{-2\varepsilon}}{n^s} \operatorname{sech}^n \varepsilon$$

converges absolutely.

2. The integral of the sum function $h(s, x)$ given in (5) over $[\varepsilon, \infty)$ has the following representation:

$$\widehat{H}(s, \varepsilon) := \int_{\varepsilon}^{\infty} h(s, x) dx = e^{-2\varepsilon} \operatorname{Li}_s(\operatorname{sech} \varepsilon), \quad (7)$$

where $\operatorname{Li}_s(z)$ is the polylogarithm function (2).

Proof. We show each point separately.

1. From Corollary 2, we have

$$\widehat{F}_n(s, \varepsilon) = \int_{\varepsilon}^{\infty} f_n(s, x) dx = \frac{e^{-2\varepsilon}}{n^s} \operatorname{sech}^n \varepsilon.$$

For $\varepsilon > 0$, we have $0 < e^{-2\varepsilon} < 1$ and $0 < \operatorname{sech} \varepsilon < 1$. Therefore,

$$|\widehat{F}_n(s, \varepsilon)| = \frac{e^{-2\varepsilon}}{n^{\operatorname{Re}(s)}} \operatorname{sech}^n \varepsilon \leq \frac{1}{n^{\operatorname{Re}(s)}}.$$

Since $\operatorname{Re}(s) > 1$, by the comparison test,

$$\sum_{n=1}^{\infty} |\widehat{F}_n(s, \varepsilon)| \leq \sum_{n=1}^{\infty} \frac{1}{n^{\operatorname{Re}(s)}} = \zeta(\operatorname{Re}(s)) < \infty,$$

where the series on the right-hand side is a convergent p -series with $p = \operatorname{Re}(s) > 1$. Thus, the series $\sum_{n=1}^{\infty} \widehat{F}_n(s, \varepsilon)$ converges absolutely.

2. For any $b > \varepsilon$, by Part 1 of Theorem 5, we can interchange the order of summation and integration to obtain

$$\int_{\varepsilon}^b h(s, x) dx = \sum_{n=1}^{\infty} \int_{\varepsilon}^b f_n(s, x) dx.$$

Taking the limit as $b \rightarrow \infty$ on both sides, we can justify the interchange of limit and summation using the absolute convergence established in Point 1. Thus, we obtain the following relationship:

$$\begin{aligned} \lim_{b \rightarrow \infty} \sum_{n=1}^{\infty} \int_{\varepsilon}^b f_n(s, x) dx &= \sum_{n=1}^{\infty} \lim_{b \rightarrow \infty} \int_{\varepsilon}^b f_n(s, x) dx \\ &= \sum_{n=1}^{\infty} \widehat{F}_n(s, \varepsilon), \end{aligned} \quad (8)$$

where equation (8) follows from Corollary 2. From Point 1, this series converges to

$$\sum_{n=1}^{\infty} \widehat{F}_n(s, \varepsilon) = \sum_{n=1}^{\infty} \frac{e^{-2\varepsilon}}{n^s} \operatorname{sech}^n \varepsilon = e^{-2\varepsilon} \sum_{n=1}^{\infty} \frac{\operatorname{sech}^n \varepsilon}{n^s} = e^{-2\varepsilon} \operatorname{Li}_s(\operatorname{sech} \varepsilon),$$

where the last equality uses the definition of the polylogarithm function. Because the right-hand side converges to a finite value as $b \rightarrow \infty$, the left-hand side must also converge. Therefore,

$$\int_{\varepsilon}^{\infty} h(s, x) dx = \lim_{b \rightarrow \infty} \int_{\varepsilon}^b h(s, x) dx = e^{-2\varepsilon} \operatorname{Li}_s(\operatorname{sech} \varepsilon).$$

This completes the proof.

Remark 8 (Broader Context of Integral Representations). The integral representation obtained in Theorem 6, which expresses $\widehat{H}(s, \varepsilon)$ (7) in terms of the polylogarithm function, belongs to a broader class of integral representations for the Riemann zeta function and its generalizations. This approach complements recent work on integral representations for the Riemann zeta function and related special functions. Sousa (2022) derived integral representations for the polylogarithm $\text{Li}_s(z)$ (2), the Hurwitz zeta function, and the Lerch transcendent at negative integers that coincide with their Abel-Plana expressions and are valid throughout $\mathbb{C}$, while Kuznetsov (2023) established a general framework for series representations of $\zeta(s)$ (1) in terms of incomplete gamma functions, hypergeometric functions, and Meijer G-functions, valid in vertical strips containing the critical line. Our framework, built on Dirichlet series with functional coefficients weighted by exponential-hyperbolic products, provides an alternative pathway to such representations, with the additional benefit of connecting directly to convergence analysis and limiting behavior.

Using Theorem 6, we can determine the antiderivative of $h(s, x)$ explicitly.

Corollary 3. For $(s, x) \in E$ (4) with $x > 0$ (i.e., excluding the boundary $x = 0$), the function

$$H(s, x) = -e^{-2x} \text{Li}_s(\text{sech } x),$$

where Li_s is the polylogarithm function (2), is an antiderivative of $h(s, x)$ from (5) on the open domain $x > 0$. That is,

$$\frac{\partial}{\partial x} H(s, x) = h(s, x) \quad \text{for all } x > 0.$$

At $x = 0$, the function $h(s, 0) = 2\zeta(s)$ is constant in x , so the antiderivative relationship holds only for $x > 0$.

Proof. We apply the product rule to the function $H(s, x)$ for $x > 0$ to calculate its partial derivative with respect to x and verify that it equals $h(s, x)$ from (5):

$$\begin{aligned} \frac{\partial}{\partial x} H(s, x) &= 2e^{-2x} \text{Li}_s(\text{sech } x) + e^{-2x} \tanh x \text{Li}_{s-1}(\text{sech } x) \\ &= 2e^{-2x} \text{Li}_s(\text{sech } x) + e^{-2x} \frac{e^x - e^{-x}}{e^x + e^{-x}} \text{Li}_{s-1}(\text{sech } x) \\ &= 2e^{-2x} \text{Li}_s(\text{sech } x) + e^{-2x} \left(\frac{2e^x}{e^x + e^{-x}} - \frac{e^x + e^{-x}}{e^x + e^{-x}} \right) \text{Li}_{s-1}(\text{sech } x) \\ &= 2e^{-2x} \text{Li}_s(\text{sech } x) + e^{-2x} (e^x \text{sech } x - 1) \text{Li}_{s-1}(\text{sech } x) \\ &= 2e^{-2x} \text{Li}_s(\text{sech } x) + e^{-x} (\text{sech } x - e^{-x}) \text{Li}_{s-1}(\text{sech } x) = h(s, x). \end{aligned}$$

This completes the proof.

Remark 9. The antiderivative formula reveals an elegant structure: For $x > 0$, the polylogarithm representation (2) of $h(s, x)$ in (5) integrates naturally to another polylogarithm expression. The combination of exponential and hyperbolic secant weights preserves the polylogarithmic form under integration. Notably, as $x \rightarrow 0^+$, both $h(s, x)$ and its antiderivative $H(s, x)$ approach expressions involving $\text{Li}_s(1) = \zeta(s)$, demonstrating the continuity of the polylogarithm structure up to the boundary.

An important special case arises when we consider the limiting behavior as s approaches the critical value $s = 1$ from the right.

Remark 10 (Limiting Behavior as $s \rightarrow 1^+$). As $s \rightarrow 1^+$, the integral representation of $h(s, x)$ (5) yields

$$\lim_{s \rightarrow 1^+} \int_{\varepsilon}^{\infty} h(s, x) dx = \lim_{s \rightarrow 1^+} e^{-2\varepsilon} \text{Li}_s(\text{sech } \varepsilon) = -e^{-2\varepsilon} \ln(1 - \text{sech } \varepsilon),$$

using the well-known special value $\text{Li}_1(z) = -\ln(1 - z)$ from (2). This limiting case connects our generalized framework to the harmonic series case, as we will see in Corollary 4 and Proposition 2. Note that although $h(s, 0) = 2\zeta(s)$ diverges as $s \rightarrow 1^+$ due to the pole of the Riemann zeta function, the integral $\widehat{H}(1, \varepsilon)$ remains finite for any $\varepsilon > 0$.

4.3 Connection to the Harmonic Series

The integral representation developed in the previous subsection allows us to investigate the limiting behavior of $h(s, x)$ (5) as $s \rightarrow 1^+$, revealing deep connections to the classical harmonic series. To rigorously justify the interchange of limit and integral mentioned in Remark 10, we first establish a general result on dominated convergence.

Lemma 4. Let $\delta > 0$ and $\varepsilon > 0$. Suppose there exists a nonnegative function $M(x)$ such that:

- (i) $|h(s, x)| \leq M(x)$ for all s with $\text{Re}(s) \in (1, 1 + \delta)$ and all $x \in [\varepsilon, \infty)$.
- (ii) The integral $\int_{\varepsilon}^{\infty} M(x) dx < \infty$ converges.

Then $\int_{\varepsilon}^{\infty} h(s, x) dx$ converges uniformly with respect to s for $\text{Re}(s) \in (1, 1 + \delta)$, where $h(s, x)$ is given in (5), and the interchange of limit and integral is justified:

$$\lim_{s \rightarrow 1^+} \int_{\varepsilon}^{\infty} h(s, x) dx = \int_{\varepsilon}^{\infty} \lim_{s \rightarrow 1^+} h(s, x) dx = \int_{\varepsilon}^{\infty} h(1, x) dx.$$

Proof. We apply the dominated convergence theorem for parameter-dependent integrals. This classical result states that if there exists an integrable majorant $M(x)$ such that $|h(s, x)| \leq M(x)$ for all s in a range and all x in the domain of integration, then the limit may be interchanged with the integral (see, e.g., Rudin (1987), Theorem 11.32, or Ahlfors (2021), Chapter 4). We verify the conditions for dominated convergence. The first step is to construct the dominating function for $h(s, x)$ given in (5). For fixed $x \geq \varepsilon$ and $\operatorname{Re}(s) \in (1, 1 + \delta)$, we take

$$M(x) := |h(1, x)| = \left| \lim_{s \rightarrow 1^+} h(s, x) \right|.$$

We then verify that this limit exists. From the polylogarithm representation of $h(s, x)$ for $x > 0$,

$$\begin{aligned} \lim_{s \rightarrow 1^+} h(s, x) &= \lim_{s \rightarrow 1^+} \left[2e^{-2x} \operatorname{Li}_s(\operatorname{sech} x) + e^{-x} (\operatorname{sech} x - e^{-x}) \operatorname{Li}_{s-1}(\operatorname{sech} x) \right] \\ &= 2e^{-2x} \operatorname{Li}_1(\operatorname{sech} x) + e^{-x} (\operatorname{sech} x - e^{-x}) \operatorname{Li}_0(\operatorname{sech} x) \\ &= -2e^{-2x} \ln(1 - \operatorname{sech} x) + e^{-x} (\operatorname{sech} x - e^{-x}) \left(\frac{\operatorname{sech} x}{1 - \operatorname{sech} x} \right) \\ &= h(1, x), \end{aligned}$$

where we used $\operatorname{Li}_1(z) = -\ln(1 - z)$ and $\operatorname{Li}_0(z) = z/(1 - z)$. This shows that $h(1, x)$ exists and is finite for all $x > 0$, with no singularity at $x = \varepsilon > 0$.

The second step is to verify the integrability of $M(x)$, for which we need to show that $\int_{\varepsilon}^{\infty} |h(1, x)| dx < \infty$. We split the integral into two parts:

$$\int_{\varepsilon}^{\infty} |h(1, x)| dx = \int_{\varepsilon}^{\xi} |h(1, x)| dx + \int_{\xi}^{\infty} |h(1, x)| dx$$

for any finite $\xi > \varepsilon$. For any value of x near ε , the function $h(1, x)$ is continuous on $[\varepsilon, \xi]$ for any finite ξ , so

$$\int_{\varepsilon}^{\xi} |h(1, x)| dx < \infty.$$

Consider the tail behavior for large x . As $x \rightarrow \infty$, we have $\operatorname{sech} x \sim 2e^{-x}$, which gives

$$-\ln(1 - \operatorname{sech} x) \sim \operatorname{sech} x \sim 2e^{-x} \quad \text{and} \quad \frac{\operatorname{sech} x}{1 - \operatorname{sech} x} \sim \operatorname{sech} x \sim 2e^{-x}.$$

Therefore, as $x \rightarrow \infty$,

$$\begin{aligned}
 |h(1, x)| &\sim 2e^{-2x} \cdot 2e^{-x} + e^{-x} \cdot (2e^{-x} - e^{-x}) \cdot 2e^{-x} \\
 &= 4e^{-3x} + 2e^{-3x} = 6e^{-3x}.
 \end{aligned}$$

Because $|h(1, x)| = O(e^{-3x})$ as $x \rightarrow \infty$, the integral

$$\int_{\xi}^{\infty} |h(1, x)| dx < \infty$$

converges. Combining both parts, we conclude that

$$\int_{\varepsilon}^{\infty} M(x) dx = \int_{\varepsilon}^{\infty} |h(1, x)| dx < \infty.$$

By the dominated convergence theorem for parameter-dependent integrals, the interchange of limit and integral is justified. This completes the proof.

By applying Lemma 4, we have rigorously established that the interchange of limit and integral is valid, allowing us to conclude the following result.

Corollary 4. *Because we have shown that $h(1, x)$ exists for $x > 0$ and verified the interchange of limit and integral, from (7) we obtain an interesting integral formula. The interchange of the limit $s \rightarrow 1^+$ with the series defining $h(s, x)$ is justified by the locally uniform convergence established in Sect. 3, which allows term-by-term passage to the limit. We have*

$$\widehat{H}(1, \varepsilon) = \int_{\varepsilon}^{\infty} h(1, x) dx = -e^{-2\varepsilon} \ln(1 - \operatorname{sech} \varepsilon), \quad \varepsilon > 0,$$

where $h(1, x)$ is a combination of harmonic-like Dirichlet series with functional coefficients:

$$\begin{aligned}
 h(1, x) &= \sum_{n=1}^{\infty} f_n(1, x) \quad (\text{with } f_n \text{ from (3)}) \\
 &= e^{-x} \sum_{n=1}^{\infty} \operatorname{sech}^{n+1} x + e^{-2x} \sum_{n=1}^{\infty} \left(\frac{2-n}{n} \right) \operatorname{sech}^n x \\
 &= e^{-x} \left(\sum_{n=1}^{\infty} \operatorname{sech}^{n+1} x - e^{-x} \sum_{n=1}^{\infty} \operatorname{sech}^n x \right) + 2e^{-2x} \sum_{n=1}^{\infty} \frac{\operatorname{sech}^n x}{n}
 \end{aligned}$$

using $\operatorname{Li}_1(z) = -\ln(1-z)$ and $\operatorname{Li}_0(z) = z/(1-z)$ from the polylogarithm definition (2).

The integral formula in Corollary 4 allows us to investigate the behavior as $\varepsilon \rightarrow 0^+$, leading to an alternative proof of a classical result.

Proposition 2 (Divergence of the Harmonic Series Via Integral Representation). *The harmonic series $\zeta(1) = \sum_{n=1}^{\infty} \frac{1}{n}$ diverges.*

Proof. From Theorem 6 and the explicit formula (7), we have for each $\varepsilon > 0$:

$$\widehat{H}(1, \varepsilon) = e^{-2\varepsilon} \sum_{n=1}^{\infty} \frac{\operatorname{sech}^n \varepsilon}{n} = -e^{-2\varepsilon} \ln(1 - \operatorname{sech} \varepsilon),$$

where the second equality follows from Corollary 4. For each fixed $\varepsilon > 0$, we have $0 < \operatorname{sech} \varepsilon < 1$, so the series converges. As $\varepsilon \rightarrow 0^+$, we have $\operatorname{sech} \varepsilon \rightarrow 1^-$ and $e^{-2\varepsilon} \rightarrow 1$. Therefore:

$$\lim_{\varepsilon \rightarrow 0^+} \widehat{H}(1, \varepsilon) = \lim_{\varepsilon \rightarrow 0^+} \left[-e^{-2\varepsilon} \ln(1 - \operatorname{sech} \varepsilon) \right] = +\infty.$$

Define $S_N := \sum_{n=1}^N \frac{1}{n}$ as the N -th partial sum of the harmonic series. For any $\varepsilon > 0$ and $N \geq 1$, because $0 < \operatorname{sech} \varepsilon < 1$, it follows that:

$$\sum_{n=1}^N \frac{\operatorname{sech}^n \varepsilon}{n} < S_N.$$

Taking the limit as $N \rightarrow \infty$, and noting that $e^{-2\varepsilon} < 1$ for $\varepsilon > 0$, we obtain:

$$\widehat{H}(1, \varepsilon) = e^{-2\varepsilon} \sum_{n=1}^{\infty} \frac{\operatorname{sech}^n \varepsilon}{n} < \sum_{n=1}^{\infty} \frac{1}{n} = \lim_{N \rightarrow \infty} S_N.$$

If the harmonic series converged, $\lim_{N \rightarrow \infty} S_N$ would be a finite value, implying that $\widehat{H}(1, \varepsilon)$ is bounded above for all $\varepsilon > 0$. However, we established that the left side approaches $+\infty$ as $\varepsilon \rightarrow 0^+$, which is a contradiction. Therefore, $\lim_{N \rightarrow \infty} S_N = +\infty$,

and the harmonic series $\zeta(1) = \sum_{n=1}^{\infty} \frac{1}{n}$ (1) diverges. This completes the proof.

The proof of Proposition 2 reveals an interesting feature of our framework: the connection between the divergence of integrals and the divergence of series.

Remark 11. This result demonstrates that although the underlying integral representation has a singularity at $x = 0$ (because $h(1, 0) = 2\zeta(1) = \infty$), we can meaningfully study the regularized behavior via $\widehat{H}(1, \varepsilon)$ for $\varepsilon > 0$. By doing so, we can precisely characterize the divergence as $\varepsilon \rightarrow 0^+$. Notably, the logarithmic

divergence of $\widehat{H}(1, \varepsilon)$ directly mirrors the well-known logarithmic growth of the partial sums of the harmonic series.

Similarly, we can also observe that although the series of integrals converges for $\varepsilon > 0$, it diverges as $\varepsilon \rightarrow 0^+$, as demonstrated in the following corollary.

Corollary 5 (Alternative Approach Via Term-by-Term Integration). *The divergence of the harmonic series can also be derived by first integrating the individual terms $f_n(s, x)$ and then summing them. From the integral representation (7), recall that*

$$\sum_{n=1}^{\infty} \widehat{F}_n(s, \varepsilon) = \sum_{n=1}^{\infty} \frac{e^{-2\varepsilon}}{n^s} \operatorname{sech}^n \varepsilon.$$

Observe that we can legitimately interchange the order of limit and summation:

$$\lim_{s \rightarrow 1^+} \sum_{n=1}^{\infty} \widehat{F}_n(s, \varepsilon) = \sum_{n=1}^{\infty} \lim_{s \rightarrow 1^+} \widehat{F}_n(s, \varepsilon).$$

This interchange is justified by dominated convergence. For $\operatorname{Re}(s) \in (1, 1 + \delta)$,

$$\left| \widehat{F}_n(s, \varepsilon) \right| = \frac{e^{-2\varepsilon}}{n^{\operatorname{Re}(s)}} \operatorname{sech}^n \varepsilon \leq e^{-2\varepsilon} \frac{\operatorname{sech}^n \varepsilon}{n},$$

and the dominating series

$$\sum_{n=1}^{\infty} e^{-2\varepsilon} \frac{\operatorname{sech}^n \varepsilon}{n} = e^{-2\varepsilon} \operatorname{Li}_1(\operatorname{sech} \varepsilon) < \infty$$

converges, where Li_1 is given by (2) with $s = 1$.

$$\lim_{s \rightarrow 1^+} \sum_{n=1}^{\infty} \widehat{F}_n(s, \varepsilon) = e^{-2\varepsilon} \sum_{n=1}^{\infty} \frac{\operatorname{sech}^n \varepsilon}{n} < \sum_{n=1}^{\infty} \frac{1}{n}.$$

Using a similar analysis as in Proposition 2, we obtain an alternative derivation showing that the harmonic series $\zeta(1)$ (1) diverges.

This corollary demonstrates the consistency of our framework that both the direct integration approach and the term-by-term integration approach lead to the same conclusion about the divergence of the harmonic series.

4.4 Special Case: $s = 2$ and the Basel Problem

The case $s = 2$ is of particular interest as it connects our generalized framework to the famous Basel problem solved by Euler in 1734. The dilogarithm function $\text{Li}_2(z)$, which corresponds to $s = 2$ in the polylogarithm definition (2), is among the most well-studied polylogarithm functions and possesses many remarkable properties and known special values. From Theorem 2, when $s = 2$, the sum function takes the following form:

$$h(2, x) = e^{-x} (\text{sech } x - e^{-x}) \text{Li}_1(\text{sech } x) + 2e^{-2x} \text{Li}_2(\text{sech } x).$$

Using $\text{Li}_1(z) = -\ln(1 - z)$, this becomes

$$h(2, x) = 2e^{-2x} \text{Li}_2(\text{sech } x) + e^{-x} (e^{-x} - \text{sech } x) \ln(1 - \text{sech } x),$$

for $x > 0$.

Proposition 3 (Connection to the Basel Problem). *The sum function $h(2, x)$ at $x = 0$ and its integral over $[\varepsilon, \infty)$ connect directly to the Basel problem.*

(1) At $x = 0$,

$$h(2, 0) = 2\zeta(2) = 2 \cdot \frac{\pi^2}{6} = \frac{\pi^2}{3}.$$

(2) The integral representation (7) from Theorem 6 with $s = 2$ yields

$$\int_{\varepsilon}^{\infty} h(2, x) dx = e^{-2\varepsilon} \text{Li}_2(\text{sech } \varepsilon).$$

(3) Taking the limit as $\varepsilon \rightarrow 0^+$ in (7) with $s = 2$ gives

$$\lim_{\varepsilon \rightarrow 0^+} \int_{\varepsilon}^{\infty} h(2, x) dx = \text{Li}_2(1) = \zeta(2) = \frac{\pi^2}{6},$$

where the special value $\text{Li}_2(1) = \zeta(2)$ follows from (2) with $z = 1$ and $s = 2$.

Proof. We verify each result individually.

(1) From Theorem 2, at $x = 0$,

$$h(2, 0) = 2\zeta(2) = 2 \sum_{n=1}^{\infty} \frac{1}{n^2} = 2 \cdot \frac{\pi^2}{6} = \frac{\pi^2}{3}.$$

(2) This follows directly from the second item of Theorem 6 with $s = 2$:

$$\widehat{H}(2, \varepsilon) = \int_{\varepsilon}^{\infty} h(2, x) dx = e^{-2\varepsilon} \operatorname{Li}_2(\operatorname{sech} \varepsilon).$$

(3) As $\varepsilon \rightarrow 0^+$, we have $e^{-2\varepsilon} \rightarrow 1$ and $\operatorname{sech} \varepsilon \rightarrow 1^-$. By continuity of Li_2 at $z = 1$, we obtain

$$\lim_{\varepsilon \rightarrow 0^+} e^{-2\varepsilon} \operatorname{Li}_2(\operatorname{sech} \varepsilon) = \operatorname{Li}_2(1) = \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

This completes the proof.

Remark 12 (Contrast with the Harmonic Case). The behavior as $\varepsilon \rightarrow 0^+$ exhibits a striking contrast between the cases $s = 1$ and $s = 2$. From (7), we observe:

Case	$h(s, 0)$	$\lim_{\varepsilon \rightarrow 0^+} \widehat{H}(s, \varepsilon)$	Behavior
$s = 1$	$2\zeta(1) = \infty$	$+\infty$	Diverges
$s = 2$	$\frac{\pi^2}{3}$	$\frac{\pi^2}{6}$	Converges

This reflects the well-known dichotomy in the convergence of p -series: $\sum_{n=1}^{\infty} \frac{1}{n^p}$ diverges for $p \leq 1$ and converges for $p > 1$. Our framework provides an integral perspective on this classical result.

Corollary 6 (Integral Formula Involving the Dilogarithm). For any $\varepsilon > 0$, we have the following explicit integral formula for $f_n(2, x)$ as defined in (3):

$$\int_{\varepsilon}^{\infty} f_n(2, x) dx = \int_{\varepsilon}^{\infty} \left[\frac{1}{n} e^{-x} \operatorname{sech}^{n+1} x + \frac{(2-n)}{n^2} e^{-2x} \operatorname{sech}^n x \right] dx = \frac{e^{-2\varepsilon}}{n^2} \operatorname{sech}^n \varepsilon.$$

Summing over all $n \geq 1$ and applying (7) with $s = 2$, we obtain

$$\int_{\varepsilon}^{\infty} h(2, x) dx = e^{-2\varepsilon} \sum_{n=1}^{\infty} \frac{\operatorname{sech}^n \varepsilon}{n^2} = e^{-2\varepsilon} \operatorname{Li}_2(\operatorname{sech} \varepsilon).$$

As $\varepsilon \rightarrow 0^+$, this yields the Basel sum:

$$\lim_{\varepsilon \rightarrow 0^+} e^{-2\varepsilon} \operatorname{Li}_2(\operatorname{sech} \varepsilon) = \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

Remark 13. This result demonstrates how our generalized framework, built upon Dirichlet series with functional coefficients involving exponential and hyperbolic secant functions, naturally recovers classical results about π and provides new integral representations for these fundamental constants. The appearance of the dilogarithm function Li_2 (the $s = 2$ case of (2)) as the natural limit object highlights the deep connections between our approach and classical analysis.

This completes our investigation of integral representations for $h(s, x)$ (5), demonstrating the utility of the interchange theorems and revealing the delicate interplay between convergence properties and limiting behaviors at critical parameter values. In the next section, we discuss some applications of the Riemann zeta function in sustainability science.

5 Applications

5.1 Harmonic Numbers for a Greener World

The study of harmonic numbers is primarily found in number theory and analysis. However, there is a rather unexplored interdisciplinary connection with potential applications to sustainability. In economics, for example, resource allocation is guided by the principle of diminishing returns, which states that as more of one input is added to a fixed amount of other inputs, the marginal output gained from each additional unit of input will eventually decrease (Forgang and Einolf 2015; Samuelson et al. 2021). Harmonic numbers naturally model situations where incremental benefits diminish.

Consider the following sustainability-related scenario, based on a classic work-rate problem, where harmonic numbers naturally arise.

Example 1 (Pollution Cleanup). Assume a major river is contaminated with a fixed amount of pollution. A community task force is organized for the cleanup, with the efficiency of each volunteer group modeled by the harmonic number. The total cleanup task is defined as one unit of pollution to be removed. The task force consists of a series of groups, indexed $k = 1, 2, 3, \dots, n$. The cleanup rate is governed by the following rule: Group 1 can remove the entire unit of pollution in 1 month. Group 2 can remove it in 2 months. In general, Group k requires k months to remove the entire unit of pollution working alone. How long will it take to clean up the river if:

1. All n groups work together.
2. The task force is infinitely staffed.

What are some sustainability implications for each case?

Solution. Let us formalize the model.

1. There are n volunteer groups working simultaneously. The individual work rate of Group k is $r_k = 1/k$ units of pollution removed per month, $k = 1, 2, 3, \dots, n$. The combined work rate is the sum of all individual rates and is given by the n -th harmonic number:

$$H_n = \sum_{k=1}^n \frac{1}{k} \quad \text{units of pollution/month.}$$

If all n groups work together, the time T_n to remove one unit of pollution is given by

$$T_n = \frac{1}{H_n} = \left(\sum_{k=1}^n \frac{1}{k} \right)^{-1} \quad \text{months.}$$

Furthermore, from the Euler–Maclaurin formula,

$$H_n \sim \ln n + \gamma,$$

where $\gamma \approx 0.5772$ is the Euler–Mascheroni constant. It follows that

$$T_n \sim \frac{1}{\ln n}.$$

As more groups are added, the cleanup time decreases at a diminishing rate proportional to the reciprocal of the logarithm.

The marginal benefit of adding group $(n+1)$ is given by the reduction in time:

$$\Delta T_n = T_n - T_{n+1} = \frac{1}{H_n} - \frac{1}{H_{n+1}}.$$

This marginal efficiency gain becomes smaller as n increases. In practice, adding the 100-th group would help much less than adding the second group, illustrating the cost-benefit trade-off in resource deployment.

2. If every possible group from $k = 1$ to ∞ is working simultaneously, then the combined work rate becomes the harmonic series that diverges to infinity:

$$\zeta(1) = \sum_{k=1}^{\infty} \frac{1}{k} = +\infty \quad \text{units of pollution/month.}$$

The time T_{∞} required to complete the cleanup is the reciprocal of the total rate:

$$T_{\infty} = \lim_{n \rightarrow \infty} T_n = \lim_{n \rightarrow \infty} \frac{1}{H_n} = \frac{1}{\zeta(1)} = 0 \quad \text{months.}$$

In this sustainability context, the mathematical conclusion is that with an infinite and ideal distribution of effort, where the slowest workers still contribute a measurable rate, the river cleanup is achieved instantaneously (in zero time, $T_\infty = 0$). This results in the paradox of infinite effort, highlighting the mathematical contrast between infinite, ideal resources and the finite, diminishing nature of real-world returns.

5.2 Generalization: Optimal Resource Allocation with Varying Efficiency Decay

The basic model outlined in Sect. 5.1 assumes that group efficiency decays linearly with group index (rate $\propto 1/k$). However, in practice, efficiency decay rates vary depending on factors such as training quality, resource availability, and coordination complexity. We can model these different scenarios by generalizing the individual work rate to

$$r_k = \frac{1}{k^s}, \quad s > 0,$$

where the parameter s controls the rate of efficiency decay. Different values of s indicate different decays:

- $s = 1$: linear decay (the harmonic case analyzed in Sect. 5.1).
- $s > 1$: faster decay.
The later groups are significantly less efficient, perhaps due to diminishing expertise or resource constraints.
- $0 < s < 1$: slower decay.
Efficiency remains relatively high even for later groups, perhaps due to effective training programs or economies of scale.

This generalization connects directly to the Riemann zeta function studied throughout this work. The combined work rate for n groups, denoted R_n , becomes

$$R_n(s) = \sum_{k=1}^n \frac{1}{k^s} \quad \text{units of pollution/month,}$$

and the cleanup time T_n is given by

$$T_n(s) = \frac{1}{R_n(s)} = \left(\sum_{k=1}^n \frac{1}{k^s} \right)^{-1} \quad \text{months.}$$

A natural question arises: How do we find the optimal number of groups? In practice, each additional group incurs costs, such as recruitment, training, and coordination, so decision-makers must balance marginal time savings against marginal costs. Suppose the cost of deploying group $(n + 1)$ is justified only if it reduces cleanup time by at least ε months (e.g., $\varepsilon = 0.01$ month ≈ 0.3 days). The marginal time reduction ΔT_n from adding group $(n + 1)$ is

$$\Delta T_n(s) = T_n(s) - T_{n+1}(s)$$

$$\begin{aligned} \Delta T_n(s) &= \frac{1}{R_n(s)} - \frac{1}{R_n(s) + (n + 1)^{-s}} \\ &= \frac{(n + 1)^{-s}}{R_n(s) [R_n(s) + (n + 1)^{-s}]}. \end{aligned}$$

We say that adding group $(n + 1)$ is *no longer cost-effective* if

$$\Delta T_n(s) < \varepsilon.$$

For sufficiently large n , we can approximate $R_n(s)$ differently depending on whether $s > 1$ or $s = 1$:

1. Case $s > 1$ (faster decay)

For $s > 1$, the series $\sum_{k=1}^{\infty} \frac{1}{k^s}$ converges to $\zeta(s)$ (1). So, for large n ,

$$R_n(s) \approx \zeta(s).$$

For large n with $s > 1$, since $(n + 1)^{-s} \ll R_n(s)$, we can approximate $R_n(s) + (n + 1)^{-s} \approx R_n(s)$. The marginal time reduction then becomes

$$\Delta T_n(s) \approx \frac{(n + 1)^{-s}}{R_n(s)^2} = \frac{1}{R_n(s)^2 (n + 1)^s} \approx \frac{1}{\zeta(s)^2 (n + 1)^s}.$$

Setting $\Delta T_n(s) = \varepsilon$ and solving for n gives the threshold:

$$n + 1 \approx \left(\frac{1}{\varepsilon \cdot \zeta(s)^2} \right)^{1/s} =: N_{\text{threshold}}(s, \varepsilon).$$

Interpretation: For $n + 1 \geq \lceil N_{\text{threshold}} \rceil$, adding another group provides negligible time savings (less than ε) and is therefore not cost-effective. The optimal deployment uses approximately $\lceil N_{\text{threshold}} \rceil - 1$ groups. Importantly, as s increases (faster efficiency decay), $N_{\text{threshold}}$ decreases, meaning fewer groups

should be deployed. The convergence of $\zeta(s)$ (1) for $s > 1$ mathematically guarantees that there exists a finite optimal number of groups.

2. Case $s = 1$ (harmonic case)

For $s = 1$, we have $R_n(1) \approx \ln n + \gamma$, and the marginal reduction becomes

$$\Delta T_n(1) \approx \frac{1}{(n+1)(\ln n + \gamma)^2} \approx \frac{1}{(n+1)(\ln n)^2},$$

where the second approximation holds for sufficiently large n with $\ln n \gg \gamma$. This decays much more slowly than the $s > 1$ case. For any fixed threshold ε , the optimal number of groups grows without bound as efficiency improves, reflecting the logarithmic divergence of the harmonic series.

Remark 14 (Mathematical Sustainability Trade-Off). This generalized model demonstrates how the mathematical distinction between convergent ($s > 1$) and divergent ($s = 1$) series directly impacts sustainability decisions:

- When $s > 1$ (fast decay), the convergence of $\zeta(s)$ (1) indicates that beyond a certain point, additional resources yield diminishing returns, and there exists a well-defined optimal allocation.
- When $s = 1$ (harmonic case), the divergence suggests that marginal benefits persist indefinitely, albeit decreasing logarithmically. Every additional group continues to provide measurable (though increasingly small) benefits.
- When $s < 1$, the series diverges even more rapidly, indicating that resource allocation strategies must be reconsidered entirely, as the mathematical framework suggests unbounded cumulative capacity.

This connection between convergence properties studied in Sects. 3 and 4 and practical resource allocation illustrates the utility of rigorous mathematical analysis for informing real-world sustainability strategies.

5.3 Emerging Applications in Sustainability

As we mentioned earlier, the Riemann zeta function plays a significant role in number theory for understanding the distribution of prime numbers (Derbyshire 2004). It has also established applications in quantum physics, probability theory, and applied statistics (Sierra 2005; Bogomolny 2007). However, its direct applications in the field of environmental science and sustainability are less conventional and still primarily emerging and theoretical.

Some potential mathematical foundations for using the Riemann zeta function include complex system modeling (Das and Kalauni 2019; Giordano et al. 2023; He et al. 2021), the dynamics of chaotic climate systems (Berry 2005; Selvam 2021), and analysis of extreme events (Fyodorov and Keating 2014; Najnudel 2018).

The connection to sustainability arises through several theoretical pathways that primarily leverage the $1/n^s$ power-law structure governing its behavior.

First, the underlying mathematical principles and analytical methods associated with the Riemann zeta function, particularly those from statistical mechanics and fractal geometry, can contribute to the general toolkit for modeling complex environmental and ecological systems (Das and Kalauni 2019; Giordano et al. 2023; He et al. 2021). These systems often exhibit self-similar patterns, where impacts or resource abundance follow a power-law distribution, e.g., species-area curves or flow distributions in resource networks. The structure of $\zeta(s)$ (1) provides a natural mathematical framework for aggregating contributions that diminish according to a power law, allowing for rigorous analysis of scale-invariant properties in sustainable resource distribution (Cohen 2020; Rafik and Humberto Salas 2024; Wilhelm and Hänggi 2003).

Second, researchers are exploring connections between the distribution of the Riemann zeta function's nontrivial zeros and the dynamics of chaotic systems. Climate systems are inherently chaotic, exhibiting sensitivity to initial conditions and irregular long-term behavior. Insights from quantum chaos theory, which has deep links to the spectral properties of the zeta zeros, may offer novel mathematical frameworks for characterizing climate variability. By viewing the climate system's dynamics through an analogous spectral lens, these techniques might help characterize the complex temporal patterns that influence climate stability and vulnerability (Berry 2005; Sakhr et al. 2003; Selvam 2021).

Third, the study of extreme values in random models related to the Riemann zeta function and random matrix theory provides theoretical frameworks that could inform the analysis of extreme weather events. The statistical properties governing the distribution's tail behavior in these models can be used to develop robust risk assessment for rare, high-impact climate events. As climate change intensifies the frequency and severity of such events, the ability to predict the bounds of these extremes becomes increasingly critical for effective sustainability planning and disaster risk mitigation (Fyodorov and Keating 2014; Najnudel 2018).

It is important to emphasize that these applications remain largely theoretical and exploratory. The development of mathematically rigorous models, whether informed by the Riemann zeta function or other advanced mathematical tools, remains crucial for improving the accuracy and reliability of climate predictions. Such advancements ultimately underpin sound sustainability policies and help define the critical boundary between systems that are mathematically convergent ($\text{Re}(s) > 1$) and those that are divergent or unsustainable.

6 Conclusion

In this chapter, we have developed a comprehensive framework for studying Dirichlet series with functional coefficients involving products of exponential and hyperbolic secant functions. Our analysis extends the classical theory of the

Riemann zeta function and harmonic series to a two-parameter setting, revealing rich mathematical structures and deep connections to established results in analysis and number theory.

Beyond the specific technical results, our central contribution is methodological. We have demonstrated that Dirichlet series with functional coefficients arising from soliton theory provide a productive framework for recovering fundamental results in number theory through alternative analytical pathways. The divergence of the harmonic series (Proposition 2) and Euler's Basel problem solution (Proposition 3) emerge naturally as limiting cases when our parameter s approaches critical values. While these results themselves are classical, our proofs are novel in their systematic use of interchange theorems, locally uniform convergence, and integral representations involving polylogarithm functions. This approach exemplifies how physics-motivated function combinations can illuminate classical mathematical structures through fresh analytical perspectives, connecting the Eulerian tradition of series manipulation with modern rigorous convergence analysis.

Beyond specific results, our study contributes several methodological insights to the analysis of Dirichlet series with functional coefficients. These include dual-parameter convergence analysis, the characterization of critical boundary behavior where uniform convergence fails, and the use of integral representations to provide alternative perspectives on series behavior. These techniques illuminate the interplay between algebraic decay (governed by the complex parameter s) and exponential decay (governed by the real parameter x through hyperbolic secant functions).

We explored potential applications of our mathematical framework to sustainability science, illustrating how harmonic numbers naturally model diminishing returns in resource allocation. The pollution cleanup example demonstrated that, while mathematical idealization with infinite resources yields instantaneous results, realistic finite-resource scenarios exhibit logarithmic efficiency decay, precisely captured by harmonic number asymptotics. This provides a concrete illustration of how abstract mathematical structures can inform practical resource management questions.

More broadly, we surveyed emerging theoretical connections between the Riemann zeta function and environmental modeling. Although these applications remain largely exploratory, the mathematical structures underlying the Riemann zeta function (including power-law distributions, scale invariance, and spectral properties) align with phenomena observed in complex environmental systems, chaotic climate dynamics, and extreme event statistics. The development of rigorous mathematical frameworks informed by advanced function theory may ultimately enhance predictive capabilities for sustainability science.

Several natural extensions of this work merit further investigation. These include analyzing Dirichlet series with functional coefficients based on other special functions exhibiting exponential decay, extending our results to complex values of the variable x to explore additional analytic properties, and developing higher dimensional generalizations with multiple parameters. Each direction promises new insights into the behavior of generalized Dirichlet series.

This work demonstrates how classical objects from analysis (harmonic series, the Riemann zeta function, and polylogarithm functions) can be profitably reexamined through the lens of Dirichlet series with functional coefficients. By introducing products of exponential and hyperbolic secant functions, we created a flexible framework that interpolates between familiar special cases while revealing new perspectives on convergence, integration, and limiting behavior. The interplay between algebraic and exponential decay mechanisms produces a rich mathematical landscape that bridges classical and contemporary analytical techniques.

As mathematics continues to develop connections with emerging application areas (from data science to climate modeling to network theory), frameworks like the one developed here may serve as bridges between classical analytical techniques and contemporary problems. The ability to rigorously characterize convergence properties, derive explicit representations, and understand limiting behavior remains essential for translating mathematical insights into practical understanding. We hope that this work will stimulate further investigations into Dirichlet series with functional coefficients and inspire new connections between classical analysis and modern applications. By demonstrating how physics-motivated functions can provide alternative analytical pathways to fundamental number-theoretic results, we aim to encourage similar cross-pollination between nonlinear dynamics, special function theory, and analytic number theory.

7 Cross-References

- ▶ [Revisiting the Notion of Integral: Calculus Versus Analysis](#)
- ▶ [The Rogers–Ramanujan Continued Fraction and the Icosahedron](#)
- ▶ [Visual, Experimental, and Computational Approaches in Theoretical Mathematics: A Contemporary Perspective](#)

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